

## ME23312 ENGINEERING THERMODYNAMICS

UNIT II – Second Law and Availability Analysis • Lecture Notes

### UNIT 2 • LECTURE NOTES

## Unit II – Second Law and Availability Analysis

Detailed lecture notes — heat engines, refrigerators and heat pumps, the second law and the Carnot cycle, Clausius inequality and entropy, the increase-of-entropy principle, available energy, availability, irreversibility and second-law efficiency, with solved numericals

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### SYLLABUS

#### UNIT II — SECOND LAW AND AVAILABILITY ANALYSIS (9 + 3 periods)

Heat Engine – Refrigerator – Heat pump. Statements of second law and their equivalence & corollaries. Carnot cycle – Reversed Carnot cycle – Performance – Clausius inequality. Concept of entropy – T-s diagram – Tds Equations – Entropy change for a pure substance. Ideal gases undergoing different processes – principle of increase in entropy. Applications of II Law. High- and low-grade energy. Availability and Irreversibility for open and closed system processes – I and II law Efficiency.

**Course Outcome CO2:** Analyze the second law of thermodynamics, accurately calculating parameters like work, heat transfer, entropy, and availability for engineering applications.

**Text:** P. K. Nag, *Engineering Thermodynamics*, 6th Edn., McGraw-Hill (Chapters 6–8). **References:** Cengel & Boles, *Thermodynamics – An Engineering Approach*, 8th Edn.; R. K. Rajput, *A Text Book of Engineering Thermodynamics*, 5th Edn.

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**Learning objectives.** After studying this unit you should be able to: (i) explain the working of a heat engine, a refrigerator and a heat pump with schematic diagrams and define the thermal efficiency and the coefficient of performance of each; (ii) state the Kelvin–Planck and Clausius statements of the second law, prove their equivalence and explain why a PMM2 is impossible; (iii) describe the Carnot cycle and the reversed Carnot cycle on  $p$ – $V$  and  $T$ – $s$  diagrams and derive their performance in terms of the reservoir temperatures; (iv) state and prove the Carnot theorem and its corollaries and use them to test claims and to analyse engine–refrigerator combinations; (v) derive the Clausius inequality and show that entropy is a property; (vi) use the  $Tds$  equations to calculate entropy changes of ideal gases and of solids, liquids and phase changes; (vii) apply the principle of increase of entropy to heat transfer, mixing and phase-change processes and compute entropy generation; (viii) distinguish high-grade from low-grade energy and available from unavailable energy; (ix) calculate the availability of closed systems and of flowing streams, the irreversibility of a process and the first- and second-law efficiencies of devices.

## 1. Introduction – why a second law is needed

The first law of thermodynamics, studied in Unit I, is a statement of the conservation of energy: in any process the energy supplied to a system equals the energy stored plus the energy delivered. It treats heat and work as completely equivalent and interchangeable, and it says nothing about whether a process can actually occur. Experience tells us otherwise. A hot cup of coffee left on a table always cools to room temperature; the reverse process – the coffee becoming hotter by absorbing heat from the cooler room – never happens, although it would not violate the first law. A paddle wheel driven by a falling weight heats the water in an insulated tank; but supplying heat to the water never makes the wheel turn and the weight rise. A gas expands to fill an evacuated vessel; it never collects itself back into one half. Fuel burns and the products cannot be un-burnt.

These observations reveal two limitations of the first law.

- **It does not indicate the direction of a process.** Both directions satisfy energy conservation, yet only one is observed. Processes in nature proceed in a definite direction and are not reversible.
- **It does not distinguish the quality of energy.** The first law treats one kilojoule of work and one kilojoule of heat as equal. In practice work can be converted completely into heat (friction, an electric heater) but heat can never be converted completely and continuously into work – a heat engine always rejects a part of the heat received. Work is a *higher-grade* energy than heat, and heat at high temperature is more valuable than the same quantity of heat at low temperature.

The **second law of thermodynamics** supplies what the first law lacks. It identifies the direction in which processes proceed, sets an upper limit to the fraction of heat that can be converted into work, defines an absolute temperature scale independent of any substance, introduces the property **entropy** as the measure of irreversibility, and – through the concepts of availability and irreversibility – quantifies the quality of energy and the loss of work potential in real processes. Unit II builds these ideas in sequence: cyclic devices, the two classical statements of the law, the Carnot cycle and theorem, the Clausius inequality, entropy, the increase-of-entropy principle and finally availability analysis.

## 2. Thermal energy reservoirs – source and sink

A **thermal energy reservoir** (or simply a reservoir) is a body of very large heat capacity that can absorb or supply an unlimited quantity of heat without any appreciable change in its temperature. The atmosphere, the oceans, large rivers and lakes are natural reservoirs; a furnace whose combustion is controlled to maintain its temperature, or a large block of ice, behaves as a reservoir in engineering analysis. Any body whose heat capacity is very large compared with the heat exchanged in the process can be treated as a reservoir – for a small electronic component the surrounding room air is a reservoir.

A reservoir that *supplies* heat is called a **source** (the furnace of a boiler, the combustion gases in an engine, the sun); a reservoir that *receives* heat is called a **sink** (the condenser cooling water, the atmosphere, a river). A reservoir is assumed to exchange heat reversibly, that is, its state changes only through the heat transferred and its temperature stays constant, so its entropy change is simply  $Q/T$ . The distinction between source and sink is by function, not by nature: the same atmosphere is a sink for a power plant and a source for a heat pump.

### 3. Heat engine, refrigerator and heat pump

The second law is stated in terms of **cyclic devices** – machines in which a working fluid executes a thermodynamic cycle repeatedly so that its state at the end of each cycle is the same as at the beginning. Three such devices are fundamental (Figure 2.1).

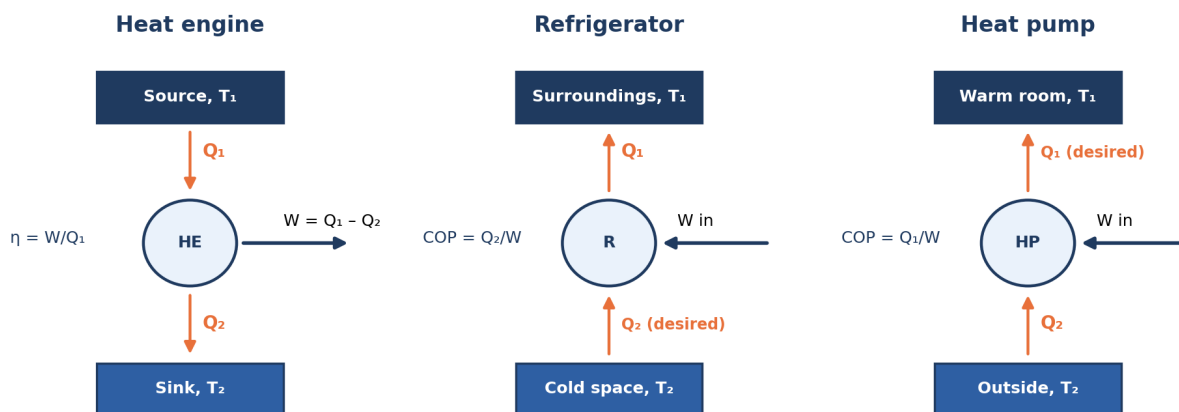


Figure 2.1 Schematic representation of a heat engine, a refrigerator and a heat pump, with the performance parameter of each.

#### 3.1 Heat engine

A **heat engine** is a cyclic device that receives heat  $Q_1$  from a high-temperature source at  $T_1$ , converts a part of it into net work  $W$  and rejects the remainder  $Q_2$  to a low-temperature sink at  $T_2$ . The working fluid (steam in a power plant, air–fuel mixture and products in an internal combustion engine, gas in a gas turbine) returns to its initial state after each cycle, so its net change of internal energy is zero and, by the first law for a cycle,

$$W_{\text{net}} = Q_1 - Q_2$$

The performance of a heat engine is measured by its **thermal efficiency**, the ratio of the desired output (net work) to the required input (heat supplied):

$$\eta_{\text{th}} = W_{\text{net}} / Q_1 = (Q_1 - Q_2) / Q_1 = 1 - Q_2/Q_1$$

Since  $Q_2$  is never zero, the efficiency of a heat engine is always less than unity. The steam power plant is the classical example: the boiler receives  $Q_1$  from the burning fuel, the turbine delivers work  $W_T$ , the condenser rejects  $Q_2$  to the cooling water, and the pump absorbs a small work  $W_P$ ; the net work  $W_T - W_P = Q_1 - Q_2$ .

#### 3.2 Refrigerator

A **refrigerator** is a cyclic device that removes heat  $Q_2$  from a low-temperature body (the cold space at  $T_2$ ) and rejects heat  $Q_1$  to a high-temperature body (the surroundings at  $T_1$ ), the transfer being made possible by a work input  $W$ . The working fluid is called the refrigerant. By the first law for the cycle,  $Q_1 = Q_2 + W$ .

The **desired effect** of a refrigerator is the heat removed from the cold space,  $Q_2$ , and its performance is measured by the **coefficient of performance**:

$$(\text{COP})_{\text{ref}} = \text{desired effect} / \text{work input} = Q_2 / W = Q_2 / (Q_1 - Q_2)$$

The COP is not called an efficiency because it can be – and normally is – greater than unity: a domestic refrigerator typically removes 2 to 4 kJ of heat for every kJ of electrical work. The rate of heat removal is often stated in **tonnes of refrigeration**; one tonne of refrigeration is the rate of heat removal needed to freeze one tonne (1000 kg) of water at 0 °C into ice at 0 °C in 24 hours,  $1 \text{ TR} = 1000 \times 334.5 / (24 \times 3600) \approx 3.87 \text{ kW}$  (the round figure 3.5 kW, based on the older 2000-lb ton, is also in common use).

### 3.3 Heat pump

A **heat pump** is the same cyclic device as a refrigerator, but its purpose is different: it delivers heat  $Q_1$  to a high-temperature body (a room to be kept warm in winter) by absorbing heat  $Q_2$  from a low-temperature body (the cold outside air or ground) with the help of a work input  $W$ . The **desired effect** is now the heat *delivered* to the warm space, so

$$(\text{COP})_{\text{HP}} = Q_1 / W = Q_1 / (Q_1 - Q_2)$$

### 3.4 Relation between the two coefficients of performance

For the same device working between the same two bodies with the same heat quantities  $Q_1$  and  $Q_2$  and the same work  $W = Q_1 - Q_2$ ,

$$(\text{COP})_{\text{HP}} = Q_1 / W = (Q_2 + W) / W = Q_2 / W + 1,$$

$$(\text{COP})_{\text{HP}} = (\text{COP})_{\text{ref}} + 1$$

Hence **the COP of a heat pump is always greater than unity**: even in the worst case, when no heat at all is absorbed from the cold body ( $Q_2 = 0$ ), the work input itself is delivered as heat to the room and the COP equals one, which is exactly what an electric resistance heater achieves. A heat pump with a COP of 4 delivers four kilojoules of heating for every kilojoule of electricity – three of them "pumped" free from the outside air – which is why heat pumps are far more economical than resistance heating.

| Device       | Purpose                | Desired effect                | Performance parameter                                     | Typical value |
|--------------|------------------------|-------------------------------|-----------------------------------------------------------|---------------|
| Heat engine  | Produce work from heat | $W_{\text{net}} = Q_1 - Q_2$  | $\eta_{\text{th}} = W_{\text{net}} / Q_1 = 1 - Q_2 / Q_1$ | 0.25 – 0.45   |
| Refrigerator | Keep a space cold      | $Q_2$ removed from cold space | $(\text{COP})_{\text{ref}} = Q_2 / W = Q_2 / (Q_1 - Q_2)$ | 2 – 4         |
| Heat pump    | Keep a space warm      | $Q_1$ delivered to warm space | $(\text{COP})_{\text{HP}} = Q_1 / W = Q_1 / (Q_1 - Q_2)$  | 3 – 5         |

**KEY POINT** A refrigerator and a heat pump are physically the same machine – a compressor, a condenser, an expansion valve and an evaporator. They differ only in what we *want* from them: the refrigerator is valued for the heat it *removes* at the low temperature, the heat pump for the heat it *delivers* at the high temperature. That is why the two COPs differ by exactly one.

## 4. Statements of the second law and their equivalence

The second law has no single mathematical formula; it is expressed as a statement of experience that has never been contradicted. Two classical statements are in use, one concerning heat engines and the other concerning refrigerators.

### 4.1 Kelvin–Planck statement

**It is impossible to construct a device that, operating in a cycle, produces no effect other than the raising of a weight (the production of work) and the exchange of heat with a single reservoir.** In other words, no heat engine can convert *all* the heat it receives into work; a part of the heat must always be rejected to a sink at a lower temperature. A heat engine must therefore exchange heat with at least two reservoirs at different temperatures. The statement does not forbid the complete conversion of heat into work in a *process* – an ideal gas expanding isothermally converts all the heat received into work – but a process is not a cycle, and the gas is left in a different state; it is the continuous, cyclic conversion that is impossible.

A hypothetical engine that would receive heat from a single reservoir and convert it entirely into work is called a **perpetual motion machine of the second kind (PMM2)**. Such a machine would not violate the first law – it creates no energy – but it would allow a ship to be propelled by extracting heat from the ocean, or a power plant to run on the heat of the atmosphere, with no fuel and no sink. The Kelvin–Planck statement may be expressed as: **a PMM2 is impossible**. (Contrast the PMM1 of Unit I, which would produce work continuously from nothing and violates the first law.)

### 4.2 Clausius statement

**It is impossible to construct a device that, operating in a cycle, produces no effect other than the transfer of heat from a cooler body to a hotter body.** Heat flows of its own accord only from a hotter to a colder body; to make it flow "uphill" a refrigerator is needed, and a refrigerator requires a work input from the surroundings. The statement does not say that heat cannot pass from a cold body to a hot one – every refrigerator does exactly that – but that it cannot do so *without some other effect*, namely the consumption of work.

### 4.3 Equivalence of the two statements

The two statements look different but are entirely equivalent: a violation of either implies a violation of the other. The proof is by contradiction and is illustrated in Figure 2.6.

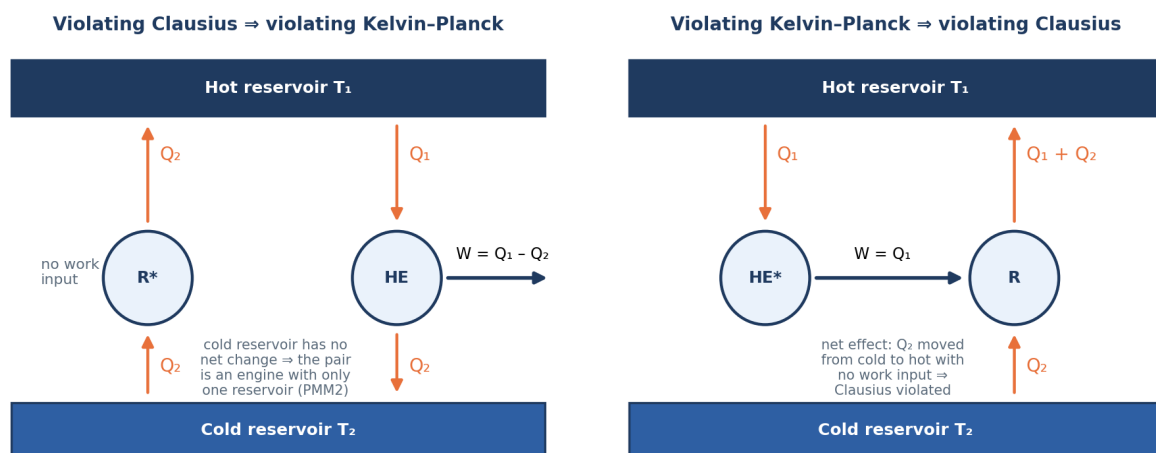


Figure 2.6 Equivalence of the Kelvin–Planck and Clausius statements: violating either statement leads to a device that violates the other.

**(a) Violation of the Clausius statement implies violation of the Kelvin–Planck statement.** Suppose a device  $R$  transfers heat  $Q_2$  from the cold reservoir at  $T_2$  to the hot reservoir at  $T_1$  with no work input (a

*Clausius violator*). Let an ordinary heat engine HE work between the same two reservoirs, receiving  $Q_1$  from the hot reservoir, delivering work  $W = Q_1 - Q_2$  and rejecting exactly  $Q_2$  to the cold reservoir. Consider R and HE together as one plant. The cold reservoir receives  $Q_2$  from HE and gives up  $Q_2$  to R\*, so it is unaffected and may be removed. The combined plant then receives a net heat  $Q_1 - Q_2$  from the hot reservoir alone and delivers an equal amount of work  $W$  – it is an engine exchanging heat with a single reservoir, a PMM2, which violates the Kelvin–Planck statement.

**(b) Violation of the Kelvin–Planck statement implies violation of the Clausius statement.** Suppose an engine HE\* receives heat  $Q_1$  from the hot reservoir and converts it completely into work  $W = Q_1$  (a Kelvin–Planck violator). Let this work drive an ordinary refrigerator R working between the same two reservoirs; R absorbs  $Q_2$  from the cold reservoir and rejects  $Q_1 + Q_2$  to the hot reservoir. For the combined plant the work is internal, the hot reservoir loses  $Q_1$  and gains  $Q_1 + Q_2$  – a net gain of  $Q_2$  – and the cold reservoir loses  $Q_2$ . The net effect is the transfer of  $Q_2$  from the cold to the hot reservoir with no external work, which violates the Clausius statement.

Since a violation of either statement leads to a violation of the other, the two statements are equivalent expressions of the same law.

#### 4.4 Corollaries of the second law

The following consequences are proved in Sections 6 and 7 and are collected here for reference: (i) no engine working between two given reservoirs can be more efficient than a reversible engine working between the same reservoirs (Carnot theorem); (ii) all reversible engines working between the same two reservoirs have the same efficiency, whatever the working substance or the details of the cycle; (iii) the efficiency of a reversible engine depends only on the two reservoir temperatures, which permits the definition of an absolute (thermodynamic) temperature scale; (iv) the COP of a reversible refrigerator or heat pump working between two reservoirs is the maximum possible and depends only on the two temperatures; (v) the entropy of an isolated system never decreases (Section 11).

### 5. Reversibility revisited – what makes a process irreversible

A **reversible process** is one which, having taken place, can be reversed so that both the system and its surroundings are restored to their initial states, leaving no trace anywhere. It is an idealisation: a reversible process must be quasi-static (infinitely slow, passing through equilibrium states) and free of all dissipative effects. Every real process is **irreversible**; the causes of irreversibility fall into two groups.

- **Mechanical irreversibility (dissipative effects):** friction between solid surfaces and in fluids, the flow of electric current through a resistance, inelastic deformation, magnetic hysteresis. In each case work is converted into internal energy (heat) and the conversion cannot be undone – heat will not spontaneously turn back into work. Paddle-wheel stirring of a fluid (Worked Example 2.7) is the classical example.
- **Lack of equilibrium (finite driving forces):** heat transfer through a finite temperature difference, unrestrained (free) expansion of a gas into a vacuum, mixing of two different gases or of two streams at different temperatures, a spontaneous chemical reaction, throttling through a valve. The process is driven by a finite difference in temperature, pressure or concentration and cannot be reversed by an infinitesimal change in the surroundings.

Irreversibility is further classified as **internal** (within the system: friction, mixing, unrestrained expansion) and **external** (at the boundary: heat transfer across a finite temperature difference between the system and its surroundings). A process that is free of internal irreversibility is called *internally reversible*; a reversible process is free of both. The importance of reversible processes is that they deliver the maximum work (or need the minimum work) and serve as the standard against which real processes are compared – the Carnot cycle of the next section is made up entirely of reversible processes.

## 6. The Carnot cycle and the reversed Carnot cycle

### 6.1 The Carnot cycle

Sadi Carnot (1824) proposed the cycle that gives the highest possible efficiency for a heat engine operating between two reservoirs. The **Carnot cycle** consists of four reversible processes executed by a working substance – here taken to be a gas in a piston–cylinder – as shown on the  $p$ – $V$  and  $T$ – $s$  planes in Figure 2.2.

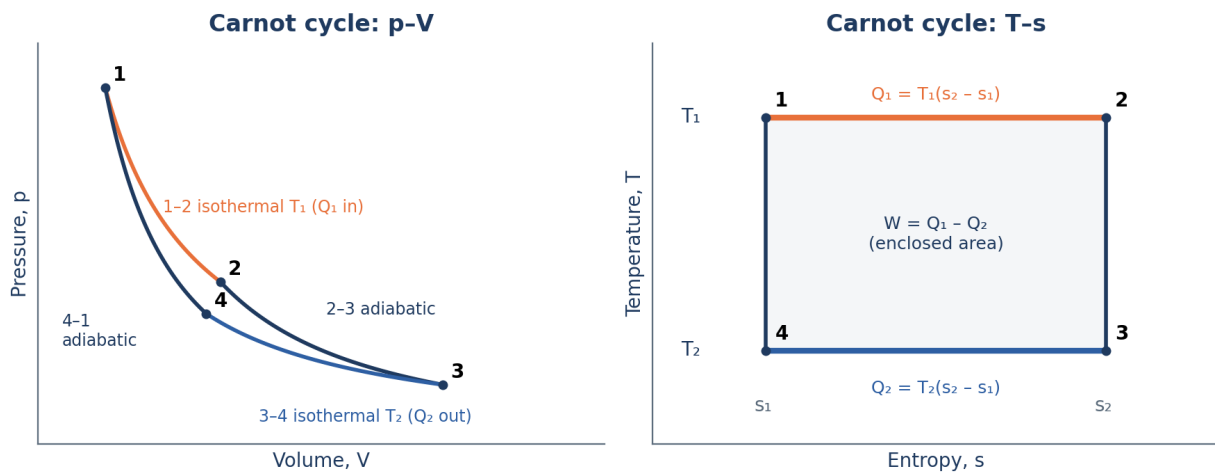


Figure 2.2 The Carnot cycle for a gas on the  $p$ – $V$  and  $T$ – $s$  planes; on the  $T$ – $s$  plane the heat quantities are the areas under the isotherms and the net work is the enclosed rectangle.

- **Process 1–2, reversible isothermal expansion at  $T_1$ .** The cylinder is placed in contact with the source at  $T_1$ ; the gas expands slowly, absorbing heat  $Q_1$  from the source while its temperature stays at  $T_1$ . For an ideal gas the internal energy is unchanged, so  $Q_1 = W_{1-2} = mRT_1 \ln(V_2/V_1)$ .
- **Process 2–3, reversible adiabatic expansion.** The cylinder is insulated and the gas continues to expand, doing work at the expense of its internal energy, until its temperature falls to  $T_2$ .
- **Process 3–4, reversible isothermal compression at  $T_2$ .** The cylinder is placed in contact with the sink at  $T_2$ ; the gas is compressed slowly, rejecting heat  $Q_2 = mRT_2 \ln(V_3/V_4)$  to the sink.
- **Process 4–1, reversible adiabatic compression.** The cylinder is insulated again and the gas is compressed until its temperature rises to  $T_1$  and it returns to state 1.

On the  $T$ – $s$  diagram the cycle is a rectangle: the two isotherms are horizontal lines and the two reversible adiabatics (isentropics) are vertical lines. The heat supplied is the area under 1–2,  $Q_1 = T_1(s_2 - s_1)$ ; the heat rejected is the area under 3–4,  $Q_2 = T_2(s_2 - s_1)$ ; and the net work, being  $Q_1 - Q_2$ , is the enclosed area.

### 6.2 Efficiency of the Carnot cycle

For the two adiabatic processes of an ideal gas,  $T_1V_2^{\gamma-1} = T_2V_3^{\gamma-1}$  and  $T_1V_1^{\gamma-1} = T_2V_4^{\gamma-1}$ ; dividing one by the other gives  $V_2/V_1 = V_3/V_4$ , so that the two logarithms in  $Q_1$  and  $Q_2$  are equal. Hence

$$\eta_{\text{Carnot}} = 1 - Q_2/Q_1 = 1 - [mRT_2 \ln(V_3/V_4)] / [mRT_1 \ln(V_2/V_1)],$$

$$\eta_{\text{Carnot}} = 1 - T_2/T_1 = (T_1 - T_2) / T_1 \quad (T \text{ in kelvin})$$

The same result follows at once from the  $T$ – $s$  diagram:  $\eta = W/Q_1 = (T_1 - T_2)(s_2 - s_1)/[T_1(s_2 - s_1)]$ . The efficiency depends only on the two reservoir temperatures, not on the working substance, and increases as  $T_1$  is raised or  $T_2$  is lowered. It can equal unity only if  $T_2 = 0 \text{ K}$  or  $T_1 \rightarrow \infty$ , neither of which is attainable; it can equal zero only if  $T_1 = T_2$ , confirming that no work can be produced from a single reservoir.

### 6.3 The reversed Carnot cycle

Since every process of the Carnot cycle is reversible, the whole cycle may be run backwards (Figure 2.3). Traversed anticlockwise, 1–2–3–4–1, it becomes a **reversed Carnot cycle** – the ideal refrigerator or heat pump: **1–2** isothermal expansion at  $T_2$  during which heat  $Q_2$  is absorbed from the cold body; **2–3** reversible adiabatic compression raising the temperature from  $T_2$  to  $T_1$ ; **3–4** isothermal compression at  $T_1$  during which heat  $Q_1$  is rejected to the hot body; **4–1** reversible adiabatic expansion lowering the temperature back to  $T_2$ . The net work input is the enclosed area,  $W = Q_1 - Q_2 = (T_1 - T_2)(s_2 - s_1)$ , and the coefficients of performance are

$$(\text{COP})_{\text{ref,Carnot}} = Q_2/(Q_1 - Q_2) = T_2/(T_1 - T_2) \quad (\text{COP})_{\text{HP,Carnot}} = Q_1/(Q_1 - Q_2) = T_1/(T_1 - T_2)$$

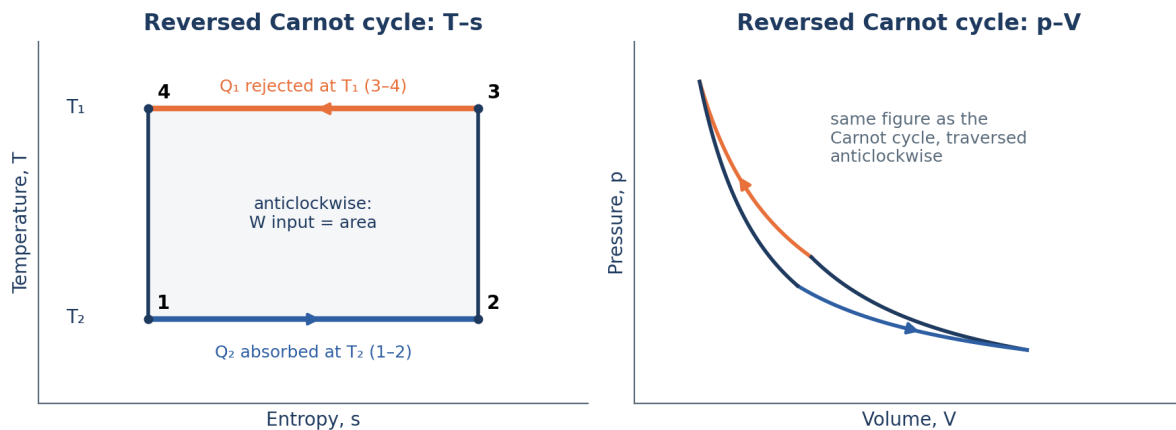


Figure 2.3 The reversed Carnot cycle – the same figure as the Carnot cycle traversed anticlockwise; the enclosed area is now the work input.

### 6.4 Why the Carnot cycle cannot be realised in practice

- The isothermal processes require heat transfer at an infinitesimal temperature difference, which means an infinitely slow process (or an infinite heat-transfer area); the adiabatic processes require a very fast process to avoid heat loss. The same piston–cylinder must therefore be extremely slow in two strokes and extremely fast in the other two, which is not feasible.
- The isothermal and adiabatic curves on the p–V diagram have nearly the same slope ( $\gamma = 1.4$  against 1 for air), so the enclosed area – the work per cycle – is very small; an engine of practical size would develop very little power.
- The cycle requires a frictionless piston, a perfectly insulating cylinder in two strokes and a perfectly conducting one in the other two, and a working substance that must be alternately heated and cooled without loss – none of which can be achieved.
- Isentropic compression of a vapour–liquid mixture (in the reversed cycle with a refrigerant) is impracticable because the compressor cannot handle two phases, and expanding the liquid in a turbine recovers very little work.

The Carnot cycle is nevertheless of the greatest importance: it establishes the *upper limit* of performance of any engine, refrigerator or heat pump working between two temperatures, and every practical cycle is judged by how closely it approaches the Carnot value.

## 7. The Carnot theorem, its corollaries and the thermodynamic temperature scale

### 7.1 Statement and proof of the Carnot theorem

**Carnot theorem:** no heat engine operating between two given thermal reservoirs can be more efficient than a reversible engine operating between the same two reservoirs.

*Proof.* Let a reversible engine R and an irreversible engine I operate between the same source at  $T_1$  and sink at  $T_2$  (Figure 2.4), and let each receive the same heat  $Q_1$  from the source. Suppose, contrary to the theorem, that  $\eta_I > \eta_R$ , so that  $W_I > W_R$  and the heats rejected satisfy  $Q_2^I = Q_1 - W_I < Q_2^R = Q_1 - W_R$ . Since R is reversible it can be run backwards as a refrigerator with the same quantities: it will then absorb  $Q_2^R$  from the sink, receive work  $W_R$  and deliver  $Q_1$  to the source. Let a part  $W_R$  of the output of I drive the reversed R. In the combined plant the source receives  $Q_1$  from R and gives  $Q_1$  to I, so it is unaffected and can be removed; the plant delivers a net work  $W_I - W_R > 0$  while exchanging heat only with the sink, from which it draws the net amount  $Q_2^R - Q_2^I = W_I - W_R$ . This is a PMM2, forbidden by the Kelvin–Planck statement. The assumption  $\eta_I > \eta_R$  is therefore false, and  $\eta_I \leq \eta_R$ .

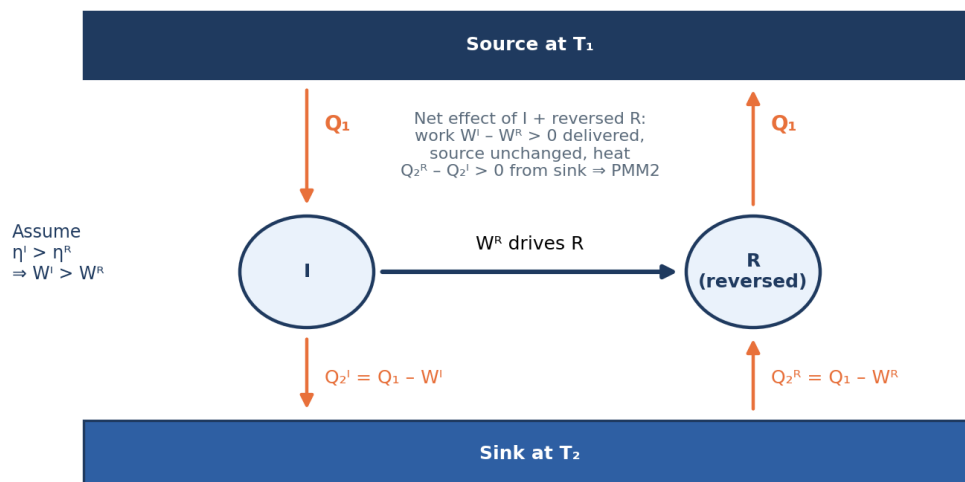


Figure 2.4 Proof of the Carnot theorem: if the irreversible engine I were more efficient than the reversible engine R, a part of its work could drive R backwards and the combination would be a PMM2.

## 7.2 Corollaries of the Carnot theorem

- **Corollary 1.** All reversible engines operating between the same two reservoirs have the same efficiency. Take two reversible engines  $R_A$  and  $R_B$ ; applying the theorem with  $R_B$  run backwards gives  $\eta_A \leq \eta_B$ , and with  $R_A$  run backwards gives  $\eta_B \leq \eta_A$ ; hence  $\eta_A = \eta_B$ . The efficiency is therefore independent of the working substance (gas, steam, any fluid) and of the mechanical details of the engine.
- **Corollary 2.** The efficiency of a reversible engine depends only on the temperatures of the two reservoirs:  $\eta_{\text{rev}} = f(T_1, T_2)$ . This is the basis of the thermodynamic temperature scale (Section 7.3).
- **Corollary 3.** No refrigerator or heat pump can have a higher COP than a reversible one operating between the same two reservoirs, and all reversible refrigerators between the same reservoirs have the same COP. The proof is identical in form to that of the theorem.

## 7.3 Thermodynamic (absolute) temperature scale

Since the efficiency of a reversible engine depends only on the reservoir temperatures, so does the ratio of the heats:  $Q_1/Q_2 = F(T_1, T_2)$ . Kelvin showed, by considering three reservoirs and reversible engines working between them in pairs, that the function must have the form  $F = \phi(T_1)/\phi(T_2)$ , and he chose the simplest function  $\phi(T) = T$ , so that

$$(Q_1/Q_2)_{\text{rev}} = T_1/T_2$$

This defines the **Kelvin (thermodynamic) temperature scale**, which is independent of the properties of any thermometric substance. The ratio of two temperatures is the ratio of the heats exchanged by a reversible engine with reservoirs at those temperatures; the size of the unit is fixed by assigning 273.16 K

to the triple point of water. With this definition the results of Section 6 hold for *any* reversible engine or refrigerator, not only one using an ideal gas:

$$\eta_{\text{rev}} = 1 - T_2/T_1 \quad (\text{COP})_{\text{ref,rev}} = T_2/(T_1 - T_2) \quad (\text{COP})_{\text{HP,rev}} = T_1/(T_1 - T_2)$$

The efficiency of any real engine is  $\eta_{\text{actual}} \leq 1 - T_2/T_1$ , and the COP of any real refrigerator is at most  $T_2/(T_1 - T_2)$ . These bounds are used constantly in engineering – to test the feasibility of a claimed performance, to find the least heat rejection or the least power for a given duty, and to rate an actual device against its ideal (Worked Examples 2.1 to 2.6). The scale also shows that  $T = 0 \text{ K}$  is the temperature at which a reversible engine would reject no heat, the lowest conceivable temperature – the absolute zero.

**KEY POINT** Always convert to kelvin before using any Carnot relation:  $T \text{ (K)} = t \text{ (}^\circ\text{C)} + 273.15$ . A temperature *difference* is the same in K and  $^\circ\text{C}$ , but a *ratio* such as  $T_2/T_1$  is meaningless in  $^\circ\text{C}$ .

## 7.4 Worked examples on heat engines, refrigerators and heat pumps

### WORKED EXAMPLE 2.1 — Least heat rejection and the test of an inventor's claim

**Problem.** A cyclic heat engine operates between a source at  $750^\circ\text{C}$  and a sink at  $25^\circ\text{C}$ . (i) Find the least possible rate of heat rejection per kW of net output. (ii) An inventor claims that an engine working between the same temperatures produces 1 kW of net work while rejecting heat at only 0.35 kW. Examine the claim. (iii) A real engine between the same limits rejects 0.7 kW per kW of net output; find its thermal efficiency and the ratio of this efficiency to the maximum possible.

**Given:**  $T_1 = 750 + 273.15 = 1023.15 \text{ K}$ ;  $T_2 = 25 + 273.15 = 298.15 \text{ K}$ ;  $W = 1 \text{ kW}$ .

**Formula:**  $\eta_{\text{max}} = 1 - T_2/T_1$ ;  $\eta = W/Q_1 = W/(W + Q_2)$ ; least  $Q_2$  corresponds to the reversible engine.

**Solution.**

(i)  $\eta_{\text{max}} = 1 - 298.15/1023.15 = 1 - 0.2914 = 0.7086$ .

For 1 kW of work the least heat input is  $Q_{1,\text{min}} = W/\eta_{\text{max}} = 1/0.7086 = 1.411 \text{ kW}$ , and the least heat rejected is  $Q_{2,\text{min}} = Q_1 - W = 1.411 - 1 = 0.411 \text{ kW}$ .

(ii) Claimed engine:  $Q_1 = W + Q_2 = 1 + 0.35 = 1.35 \text{ kW}$ ;  $\eta_{\text{claimed}} = 1/1.35 = 0.741$ . This exceeds  $\eta_{\text{max}} = 0.709$ , so the claim violates the Carnot principle (it would reject less than the minimum 0.411 kW). The claim is false.

(iii) Real engine:  $Q_1 = 1 + 0.7 = 1.7 \text{ kW}$ ;  $\eta = 1/1.7 = 0.588$ ; ratio  $\eta/\eta_{\text{max}} = 0.588/0.709 = 0.830$ .

**ANSWER** (i) Least heat rejection 0.411 kW per kW of work; (ii) the claimed efficiency of 74.1 % exceeds the Carnot limit of 70.9 % – the engine is impossible; (iii)  $\eta = 58.8 \%$ , which is 83.0 % of the maximum possible.

### WORKED EXAMPLE 2.2 — Two reversible engines in series

**Problem.** Two reversible heat engines A and B are arranged in series, A rejecting heat directly to B. Engine A receives 250 kJ from a source at  $450^\circ\text{C}$  and engine B rejects heat to a sink at  $10^\circ\text{C}$ . If the work output of A is twice that of B, determine (i) the intermediate temperature between A and B, (ii) the efficiency of each engine and (iii) the heat rejected to the sink.

**Given:**  $T_1 = 450 + 273.15 = 723.15 \text{ K}$ ;  $T_3 = 10 + 273.15 = 283.15 \text{ K}$ ;  $Q_1 = 250 \text{ kJ}$ ;  $W_A = 2W_B$ . Let the intermediate temperature be  $T_2$  (Figure 2.7, left).

**Formula:** For reversible engines  $Q_1/T_1 = Q_2/T_2 = Q_3/T_3$ ;  $W_A = Q_1 - Q_2$ ;  $W_B = Q_2 - Q_3$ ;  $\eta = 1 - T_{\text{low}}/T_{\text{high}}$ .

**Solution.**

(i) With  $Q_2 = Q_1 T_2 / T_1$  and  $Q_3 = Q_1 T_3 / T_1$ :  $W_A = Q_1(T_1 - T_2) / T_1$  and  $W_B = Q_1(T_2 - T_3) / T_1$ .

$W_A = 2W_B \Rightarrow T_1 - T_2 = 2(T_2 - T_3) \Rightarrow T_2 = (T_1 + 2T_3) / 3 = (723.15 + 566.3) / 3 = 429.8 \text{ K} = 156.7 \text{ }^\circ\text{C}$ .

(ii)  $\eta_A = 1 - T_2 / T_1 = 1 - 429.8 / 723.15 = 0.406$ ;  $\eta_B = 1 - T_3 / T_2 = 1 - 283.15 / 429.8 = 0.341$ .

(iii)  $W_A = \eta_A Q_1 = 0.406 \times 250 = 101.4 \text{ kJ}$ ;  $Q_2 = 250 - 101.4 = 148.6 \text{ kJ}$ ;  $W_B = \eta_B Q_2 = 0.341 \times 148.6 = 50.7 \text{ kJ}$  ( $= W_A / 2 \checkmark$ );  $Q_3 = Q_2 - W_B = 148.6 - 50.7 = 97.9 \text{ kJ}$ . Check:  $Q_3 = Q_1 T_3 / T_1 = 250 \times 283.15 / 723.15 = 97.9 \text{ kJ} \checkmark$ .

**ANSWER** (i)  $T_2 = 429.8 \text{ K}$  ( $156.7 \text{ }^\circ\text{C}$ ); (ii)  $\eta_A = 40.6 \%$ ,  $\eta_B = 34.1 \%$ ; (iii)  $Q_3 = 97.9 \text{ kJ}$ .

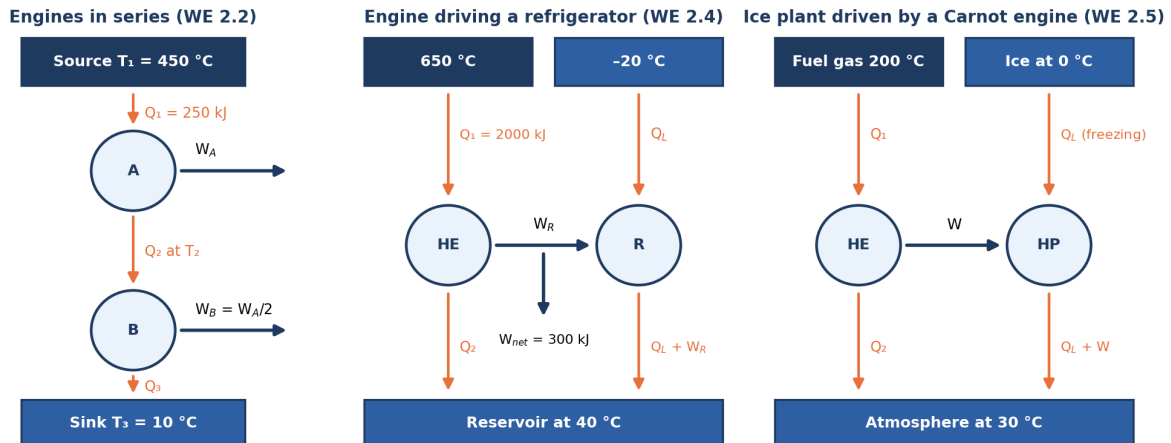


Figure 2.7 Arrangements analysed in Worked Examples 2.2, 2.4 and 2.5.

### WORKED EXAMPLE 2.3 — Heat pump for winter heating and summer cooling

**Problem.** A reversible heat pump keeps a house at  $22 \text{ }^\circ\text{C}$  in winter and, reversed, cools it to  $22 \text{ }^\circ\text{C}$  in summer. Heat transfer through the walls and roof is  $0.6 \text{ kW}$  per kelvin of inside–outside temperature difference. (i) Find the minimum power to drive the heat pump when the outside temperature in winter is  $2 \text{ }^\circ\text{C}$ . (ii) With the same power input, find the highest outside temperature in summer at which  $22 \text{ }^\circ\text{C}$  can be maintained inside.

**Given:**  $T_i = 22 + 273.15 = 295.15 \text{ K}$ ; winter  $T_o = 2 + 273.15 = 275.15 \text{ K}$ ; heat leakage rate  $k = 0.6 \text{ kW/K}$ .

**Formula:** Heating load  $Q = k(T_i - T_o)$ ; winter  $(\text{COP})_{\text{HP}} = T_i / (T_i - T_o)$ ,  $W = Q / (\text{COP})_{\text{HP}}$ ; summer  $(\text{COP})_{\text{ref}} = T_i / (T_o - T_i)$ , cooling load  $Q = k(T_o - T_i)$ .

**Solution.**

(i) Winter: heat to be delivered  $Q_1 = 0.6 \times (22 - 2) = 12 \text{ kW}$ .

$(\text{COP})_{\text{HP}} = 295.15 / (295.15 - 275.15) = 295.15 / 20 = 14.76$ .

$W_{\min} = Q_1 / \text{COP} = 12 / 14.76 = 0.813 \text{ kW}$ .

(ii) Summer: let the outside temperature be  $T_o$  and  $\Delta T = T_o - T_i$ . Heat leaking in  $= 0.6 \Delta T$ ;  $(\text{COP})_{\text{ref}} = T_i / \Delta T$ ; work needed  $W = 0.6 \Delta T \times \Delta T / T_i = 0.6 \Delta T^2 / 295.15$ .

Setting  $W = 0.813 \text{ kW}$ :  $\Delta T^2 = 0.813 \times 295.15 / 0.6 = 400 \Rightarrow \Delta T = 20 \text{ K}$ . (In both seasons  $W = k \Delta T^2 / T_i$ , because the inside temperature  $T_i$  is the numerator of the heat-pump COP in winter and of the refrigerator COP in summer; the same power therefore handles the same temperature difference in either direction.)

$T_{o,\max} = 22 + 20 = 42 \text{ }^\circ\text{C}$ .

**ANSWER** (i) Minimum power  $0.813 \text{ kW}$ ; (ii) the house can be held at  $22 \text{ }^\circ\text{C}$  for outside temperatures up to  $42 \text{ }^\circ\text{C}$ .

**WORKED EXAMPLE 2.4 — Reversible engine driving a reversible refrigerator**

**Problem.** A reversible heat engine operates between reservoirs at 650 °C and 40 °C and drives a reversible refrigerator working between 40 °C and –20 °C. The engine receives 2000 kJ and the net work output of the combined plant is 300 kJ. (i) Find the heat absorbed by the refrigerant and the net heat rejected to the 40 °C reservoir. (ii) Repeat the analysis if the engine efficiency and the refrigerator COP are each 50 % of their maximum values, and compare.

**Given:**  $T_1 = 923.15 \text{ K}$ ,  $T_2 = 313.15 \text{ K}$ ,  $T_L = 253.15 \text{ K}$ ;  $Q_1 = 2000 \text{ kJ}$ ;  $W_{\text{net}} = 300 \text{ kJ}$  (Figure 2.7, middle).

**Formula:**  $\eta_E = 1 - T_2/T_1$ ;  $W_E = \eta_E Q_1$ ;  $Q_2 = Q_1 - W_E$ ;  $W_R = W_E - W_{\text{net}}$ ;  $(\text{COP})_R = T_L/(T_2 - T_L)$ ;  $Q_L = \text{COP} \times W_R$ ;  $Q_H = Q_L + W_R$ ; net heat to the 40 °C reservoir =  $Q_2 + Q_H$ .

**Solution.**

(i) Reversible plant:  $\eta_E = 1 - 313.15/923.15 = 0.6608$ ;  $W_E = 0.6608 \times 2000 = 1321.6 \text{ kJ}$ ;  $Q_2 = 2000 - 1321.6 = 678.4 \text{ kJ}$ .

Work to the refrigerator  $W_R = 1321.6 - 300 = 1021.6 \text{ kJ}$ .

$(\text{COP})_R = 253.15/(313.15 - 253.15) = 253.15/60 = 4.219$ ;  $Q_L = 4.219 \times 1021.6 = 4310 \text{ kJ}$ ;  $Q_H = 4310 + 1021.6 = 5332 \text{ kJ}$ .

Net heat to the 40 °C reservoir =  $678.4 + 5332 = 6010 \text{ kJ}$ .

(ii) Actual plant:  $\eta_E = 0.5 \times 0.6608 = 0.3304$ ;  $W_E = 660.8 \text{ kJ}$ ;  $Q_2 = 1339.2 \text{ kJ}$ ;  $W_R = 660.8 - 300 = 360.8 \text{ kJ}$ .

$(\text{COP})_R = 0.5 \times 4.219 = 2.110$ ;  $Q_L = 2.110 \times 360.8 = 761 \text{ kJ}$ ;  $Q_H = 761 + 360.8 = 1122 \text{ kJ}$ .

Net heat to the 40 °C reservoir =  $1339.2 + 1122 = 2461 \text{ kJ}$ .

**ANSWER** (i) Reversible:  $Q_L = 4310 \text{ kJ}$ , net heat to the 40 °C reservoir 6010 kJ. (ii) At 50 % of ideal performance the refrigerating effect falls to 761 kJ – less than one-fifth of the ideal value – because both the work available to the refrigerator and its COP are halved; the heat to the 40 °C reservoir falls to 2461 kJ. Energy is conserved in both cases ( $2000 + Q_L = 300 + \text{net heat rejected}$ ).

**WORKED EXAMPLE 2.5 — Ice plant driven by a Carnot engine**

**Problem.** An ice plant working on a reversed Carnot cycle produces 12 tonnes of ice per day from water at 0 °C, the ice being kept at 0 °C, and rejects heat to the atmosphere at 30 °C. The heat pump is driven by a Carnot engine that receives heat from a source at 200 °C, maintained by burning a fuel of calorific value 42000 kJ/kg, and rejects heat to the atmosphere. Find (i) the power developed by the engine and (ii) the fuel consumed per hour. Enthalpy of fusion of ice = 334.5 kJ/kg.

**Given:** Ice rate = 12000 kg/day;  $T_{\text{ice}} = 273.15 \text{ K}$ ;  $T_0 = 30 + 273.15 = 303.15 \text{ K}$ ;  $T_s = 200 + 273.15 = 473.15 \text{ K}$ ;  $\text{CV} = 42000 \text{ kJ/kg}$ ;  $h_{\text{sf}} = 334.5 \text{ kJ/kg}$  (Figure 2.7, right).

**Formula:**  $Q_L = \dot{m}_{\text{ice}} h_{\text{sf}}$ ;  $(\text{COP})_{\text{ref}} = T_{\text{ice}}/(T_0 - T_{\text{ice}})$ ;  $W = Q_L/\text{COP}$ ;  $\eta_E = 1 - T_0/T_s$ ;  $Q_1 = W/\eta_E$ ; fuel rate =  $Q_1/\text{CV}$ .

**Solution.**

$\dot{m}_{\text{ice}} = 12000/(24 \times 3600) = 0.1389 \text{ kg/s}$ ;  $Q_L = 0.1389 \times 334.5 = 46.46 \text{ kW}$ .

$(\text{COP})_{\text{ref}} = 273.15/(303.15 - 273.15) = 273.15/30 = 9.105$ .

(i)  $W = 46.46/9.105 = 5.10 \text{ kW}$  – the power the engine must develop.

$\eta_E = 1 - 303.15/473.15 = 0.3593$ ;  $Q_1 = 5.10/0.3593 = 14.20 \text{ kW}$ .

(ii) Fuel rate =  $14.20/42000 = 3.381 \times 10^{-4} \text{ kg/s} = 3.381 \times 10^{-4} \times 3600 = 1.217 \text{ kg/h}$ .

**ANSWER** (i) Power developed by the engine 5.10 kW; (ii) fuel consumption 1.217 kg/h. (The atmosphere receives  $46.46 + 5.10 = 51.56$  kW from the heat pump and  $14.20 - 5.10 = 9.10$  kW from the engine.)

**WORKED EXAMPLE 2.6 — Domestic freezer – least power, actual COP and effect of a lower temperature**

**Problem.** A domestic freezer holds  $-18^\circ\text{C}$  while the room is at  $28^\circ\text{C}$ ; heat leaks in continuously at 1.5 kW. (i) Find the least power needed to remove this heat. (ii) The compressor actually draws 0.6 kW: find the actual COP, the heat rejected to the room and the ratio of actual to maximum COP. (iii) How does the least power change if the freezer is held at  $-28^\circ\text{C}$  with the same leakage?

**Given:**  $T_L = -18 + 273.15 = 255.15$  K;  $T_H = 28 + 273.15 = 301.15$  K;  $Q_L = 1.5$  kW;  $W_{\text{actual}} = 0.6$  kW.

**Formula:**  $(\text{COP})_{\text{max}} = T_L/(T_H - T_L)$ ;  $W_{\text{min}} = Q_L/(\text{COP})_{\text{max}}$ ;  $(\text{COP})_{\text{actual}} = Q_L/W_{\text{actual}}$ ;  $Q_H = Q_L + W$ .

**Solution.**

(i)  $(\text{COP})_{\text{max}} = 255.15/(301.15 - 255.15) = 255.15/46 = 5.547$ ;  $W_{\text{min}} = 1.5/5.547 = 0.270$  kW.

(ii)  $(\text{COP})_{\text{actual}} = 1.5/0.6 = 2.5$ ; heat rejected  $Q_H = 1.5 + 0.6 = 2.1$  kW; ratio  $= 2.5/5.547 = 0.451$  – the freezer achieves 45 % of the Carnot COP.

(iii) At  $-28^\circ\text{C}$ :  $T_L = 245.15$  K;  $(\text{COP})_{\text{max}} = 245.15/(301.15 - 245.15) = 245.15/56 = 4.378$ ;  $W_{\text{min}} = 1.5/4.378 = 0.343$  kW, an increase of  $(0.343 - 0.270)/0.270 = 26.7$  %.

**ANSWER** (i) 0.270 kW; (ii) COP = 2.5,  $Q_H = 2.1$  kW, 45.1 % of the maximum COP; (iii) lowering the freezer temperature by 10 K raises the least power to 0.343 kW (+26.7 %) – colder storage is always more expensive, and in practice the leakage would also increase.

## 8. The Clausius inequality

The Carnot relations of Section 7 concern heat exchanged with two reservoirs at fixed temperatures. Clausius generalised them to *any* cycle in which heat is exchanged at *any* number of temperatures, and the result – the Clausius inequality – is the doorway to the property entropy.

### 8.1 Derivation

Adopt the usual sign convention: heat received by the system is positive, heat rejected is negative. For a **reversible engine** working between reservoirs at  $T_1$  and  $T_2$ , Section 7.3 gives  $Q_1/T_1 = Q_2/T_2$ , where  $Q_1$  is received and  $Q_2$  rejected. With the sign convention the rejected heat is  $-Q_2$ , so the algebraic sum of  $Q/T$  around the cycle is

$$Q_1/T_1 + (-Q_2)/T_2 = 0.$$

For an **irreversible engine** between the same reservoirs, the Carnot theorem gives  $\eta_i < \eta_R$ , i.e.  $1 - Q_2/Q_1 < 1 - T_2/T_1$ , so  $Q_2/Q_1 > T_2/T_1$  and  $Q_2/T_2 > Q_1/T_1$ . Hence

$$Q_1/T_1 + (-Q_2)/T_2 < 0.$$

Now consider an arbitrary cycle in which the system receives and rejects heat at continuously varying temperatures. Divide the cycle by a large number of closely spaced reversible adiabatics into elementary strips; each strip, closed by two short reversible isotherms, is an elementary Carnot cycle receiving  $\delta Q_1$  at  $T_1$  and rejecting  $\delta Q_2$  at  $T_2$ , and the isotherm segments of neighbouring strips cancel in pairs. Summing the result above over all the strips and passing to the limit gives, for the whole cycle,

$$\oint \delta Q/T \leq 0 \quad (\text{Clausius inequality: } = 0 \text{ for a reversible cycle, } < 0 \text{ for an irreversible cycle})$$

where  $T$  is the absolute temperature at the part of the boundary through which  $\delta Q$  crosses. The cyclic integral of  $\delta Q/T$  can never be positive: a cycle for which  $\oint \delta Q/T > 0$  is impossible, because it would contain an engine more efficient than a Carnot engine. The inequality is thus a mathematical statement of the second law for cycles; the quantity  $\oint \delta Q/T$  is a measure of the irreversibility of the cycle, and its value  $-\oint \delta Q/T$  is proportional to the entropy generated in each cycle.

## 8.2 Illustration

A heat engine receives 1000 kJ from a source at 800 K and rejects 500 kJ to a sink at 300 K. Here  $\oint \delta Q/T = 1000/800 - 500/300 = 1.25 - 1.667 = -0.417 \text{ kJ/K} < 0$ : the cycle is irreversible but possible (its efficiency is 50 % against the Carnot 62.5 %). If it rejected only 300 kJ,  $\oint \delta Q/T = 1.25 - 1.0 = +0.25 \text{ kJ/K} > 0$  and the cycle is impossible. If it rejected exactly 375 kJ,  $\oint \delta Q/T = 1.25 - 1.25 = 0$  and the cycle is reversible.

## 9. Entropy – a property of the system; the T–s diagram

### 9.1 Entropy is a property

Let a system change from state 1 to state 2 along a reversible path A and return to state 1 along another reversible path B (Figure 2.5, left, shows one such path). The cycle A–B is reversible, so by the Clausius inequality

$$\oint \delta Q/T = \int_A \delta Q/T (1 \rightarrow 2) + \int_B \delta Q/T (2 \rightarrow 1) = 0.$$

If the return is made instead along a third reversible path C, similarly  $\int_A \delta Q/T (1 \rightarrow 2) + \int_C \delta Q/T (2 \rightarrow 1) = 0$ . Subtracting,  $\int_B \delta Q/T (2 \rightarrow 1) = \int_C \delta Q/T (2 \rightarrow 1)$ : the integral of  $\delta Q/T$  along a reversible path between two states is **independent of the path** and depends only on the end states. It therefore represents the change of a property. Clausius named this property **entropy**,  $S$ :

$$dS = (\delta Q/T)_{\text{rev}} \quad S_2 - S_1 = \int_1^2 (\delta Q/T)_{\text{rev}} \quad [\text{kJ/K}]$$

Entropy is an extensive property; the specific entropy  $s = S/m$  has the unit kJ/kg K. Only *changes* of entropy are defined by this relation; the third law fixes the zero (the entropy of a perfect crystal at 0 K is zero), but for engineering work an arbitrary datum is used – for steam,  $s = 0$  for saturated liquid at the triple point. Being a property, the entropy change between two states is the same whether the process is reversible or irreversible; to evaluate it for an irreversible process we simply *imagine* any convenient reversible path between the same end states and integrate  $\delta Q/T$  along it.

### 9.2 The T–s diagram: heat transfer as an area

Since  $\delta Q_{\text{rev}} = T dS$ , the heat transferred in a reversible process is  $Q_{1-2} = \int T dS$ , the **area under the process curve on the temperature–entropy diagram** (Figure 2.5, left) – exactly as the work of a quasi-static process is the area under the curve on the  $p$ – $V$  diagram. If the entropy increases the system receives heat; if it decreases the system rejects heat. A reversible adiabatic process ( $\delta Q = 0$ ) is one of constant entropy – an **isentropic** process, a vertical line on the T–s diagram; a reversible isothermal process is a horizontal line, and the heat transferred is  $Q = T(S_2 - S_1)$ . For a cycle the enclosed area on the T–s diagram is the net heat, which equals the net work. The Carnot cycle is a rectangle, and the diagram makes its efficiency and the effect of raising  $T_1$  or lowering  $T_2$  immediately visible.

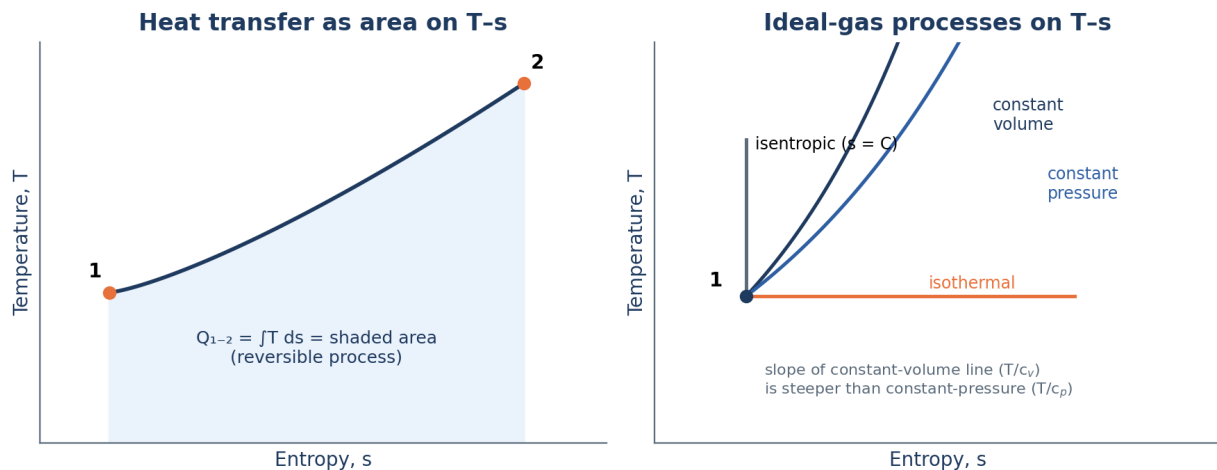


Figure 2.5 (Left) Heat transfer in a reversible process is the area under the curve on the  $T$ - $s$  diagram. (Right) Constant-volume, constant-pressure, isothermal and isentropic lines of an ideal gas on the  $T$ - $s$  plane.

For an ideal gas the slope of a constant-volume line on the  $T$ - $s$  diagram is  $(\partial T/\partial s)_v = T/c_v$  and that of a constant-pressure line is  $(\partial T/\partial s)_p = T/c_p$ ; since  $c_p > c_v$  the constant-volume line is the steeper of the two (Figure 2.5, right). These slopes follow directly from the  $Tds$  equations of the next section.

**KEY POINT** The equation  $dS = \delta Q/T$  holds *only* for a reversible process. For an irreversible process  $dS > \delta Q/T$  (Section 11): entropy is created inside the system by irreversibility, in addition to the entropy carried in by heat. Entropy is transferred only by heat, never by work – work is "entropy-free" energy, which is the deepest reason why it is the higher grade.

## 10. The $Tds$ equations and entropy change of ideal gases and pure substances

### 10.1 The two $Tds$ equations

For a closed system undergoing a reversible process the first law gives  $\delta Q_{\text{rev}} = dU + p dv$ , and the definition of entropy gives  $\delta Q_{\text{rev}} = T dS$ . Combining them,

$$T dS = dU + p dv \quad \text{or, per unit mass, } T ds = du + p dv \quad (\text{first } Tds \text{ equation})$$

Since  $H = U + pV$ ,  $dH = dU + p dv + V dp$ , so  $dU + p dv = dH - V dp$  and

$$T dS = dH - V dp \quad \text{or } T ds = dh - v dp \quad (\text{second } Tds \text{ equation})$$

Although they were derived for a reversible process, both equations contain only *properties* ( $T$ ,  $s$ ,  $u$ ,  $h$ ,  $p$ ,  $v$ ) and are therefore true for **any** process, reversible or irreversible, between two equilibrium states – they are relations among properties, not statements about a path. They are the working tools for all entropy calculations:  $ds = du/T + (p/T) dv$  and  $ds = dh/T - (v/T) dp$ , with the equation of state supplying  $p/T$  or  $v/T$ .

### 10.2 Entropy change of an ideal gas

For an ideal gas  $du = c_v dT$ ,  $dh = c_p dT$ ,  $p/T = R/v$  and  $v/T = R/p$ . The first  $Tds$  equation gives  $ds = c_v dT/T + R dv/v$  and the second  $ds = c_p dT/T - R dp/p$ . Integrating with constant specific heats:

$$s_2 - s_1 = c_v \ln(T_2/T_1) + R \ln(v_2/v_1) = c_p \ln(T_2/T_1) - R \ln(p_2/p_1) = c_v \ln(p_2/p_1) + c_p \ln(v_2/v_1)$$

(The third form follows from the first by substituting  $T_2/T_1 = p_2 v_2 / (p_1 v_1)$  and using  $c_v + R = c_p$ .) Multiplying by the mass  $m$  gives  $S_2 - S_1$ . Applied to the standard processes of an ideal gas of mass  $m$ :

| Process                           | Law             | Entropy change $S_2 - S_1$                                       | Heat transfer $Q_{1-2}$ (reversible)                  |
|-----------------------------------|-----------------|------------------------------------------------------------------|-------------------------------------------------------|
| Constant volume                   | $v = C$         | $m c_v \ln(T_2/T_1) = m c_v \ln(p_2/p_1)$                        | $m c_v (T_2 - T_1)$                                   |
| Constant pressure                 | $p = C$         | $m c_p \ln(T_2/T_1) = m c_p \ln(V_2/V_1)$                        | $m c_p (T_2 - T_1)$                                   |
| Isothermal                        | $T = C$         | $m R \ln(V_2/V_1) = m R \ln(p_1/p_2) = Q/T$                      | $m R T \ln(V_2/V_1)$                                  |
| Reversible adiabatic (isentropic) | $pV^\gamma = C$ | 0                                                                | 0                                                     |
| Polytropic                        | $pV^n = C$      | $m c_v [(n - \gamma)/(n - 1)] \ln(T_2/T_1) = m c_n \ln(T_2/T_1)$ | $m c_n (T_2 - T_1)$ , $c_n = c_v(n - \gamma)/(n - 1)$ |

The polytropic result is obtained from  $s_2 - s_1 = c_v \ln(T_2/T_1) + R \ln(v_2/v_1)$  with  $Tv^{n-1} = C$ , i.e.  $\ln(v_2/v_1) = -\ln(T_2/T_1)/(n - 1)$ , and  $R = c_v(\gamma - 1)$ . For  $1 < n < \gamma$  (the usual compression with cooling)  $c_n$  is negative: the temperature rises while heat is rejected and the entropy falls. On the  $T$ - $s$  diagram (Figure 2.5, right) the isothermal line is horizontal, the isentropic vertical, and the constant-volume and constant-pressure lines are exponential curves  $T = T_1 \exp[(s - s_1)/c_v]$  and  $T_1 \exp[(s - s_1)/c_p]$ .

### 10.3 Entropy change of solids, liquids and phase changes

For a solid or a liquid the specific volume is nearly constant, so the  $p dv$  term vanishes and  $c_p \approx c_v = c$ ; then  $ds = c dT/T$  and, if  $c$  is constant,

$$S_2 - S_1 = m c \ln(T_2/T_1) \quad (\text{solid or liquid, constant specific heat})$$

During a **phase change at constant pressure** (melting, evaporation) the temperature also stays constant, so the entropy change is the latent heat divided by the absolute temperature at which it is absorbed:

$$S_2 - S_1 = m h_{fg}/T_{\text{sat}} \quad (\text{evaporation}) \quad S_2 - S_1 = m h_{sf}/T_{\text{melt}} \quad (\text{melting})$$

For a **vapour** whose specific heat varies with temperature,  $c_p = a + bT + cT^2$ , the second  $Tds$  equation at constant pressure gives  $ds = c_p dT/T$  and  $s_2 - s_1 = a \ln(T_2/T_1) + b(T_2 - T_1) + (c/2)(T_2^2 - T_1^2)$ . Worked Example 2.8 combines all three results for water heated from a cold liquid to superheated steam. In practice the entropy of a pure substance is read from **property tables** (Unit III): for a wet mixture  $s = s_f + x s_{fg}$ , for a compressed liquid  $s \approx s_f$  at the same temperature, and for superheated vapour  $s$  is tabulated against  $p$  and  $T$ ; the entropy of saturated water and steam is measured from the datum  $s = 0$  at the triple point.

## 11. Entropy change in irreversible processes – the principle of increase of entropy

### 11.1 Entropy change of a system in an irreversible process

Let a system go from state 1 to state 2 by an irreversible process I and return to state 1 by a reversible process R (any reversible path). The cycle I + R is irreversible, so the Clausius inequality gives

$$\int_I \delta Q/T (1 \rightarrow 2) + \int_R \delta Q/T (2 \rightarrow 1) < 0.$$

The second integral, being along a reversible path, equals  $S_1 - S_2$ . Hence  $\int_I \delta Q/T (1 \rightarrow 2) < S_2 - S_1$ , or in differential form

$$dS \geq \delta Q/T \quad (= \text{for a reversible process, } > \text{ for an irreversible process})$$

The entropy change of a system is *at least* equal to the entropy transferred into it with heat,  $\int \delta Q/T$ ; the excess is the entropy **generated** (produced) within the system by the irreversibilities of the process:

$$S_2 - S_1 = \int_1^2 \delta Q/T + S_{\text{gen}} \quad \text{with } S_{\text{gen}} \geq 0$$

$S_{\text{gen}}$  is not a property – it depends on the path – and it is never negative; it is zero only for a reversible process. Note that the entropy of a system *can* decrease (when it rejects heat,  $\int \delta Q/T$  is negative and may outweigh  $S_{\text{gen}}$ ), but only by transferring at least as much entropy to something else.

## 11.2 The principle of increase of entropy

For an **isolated system** there is no heat transfer,  $\delta Q = 0$ , so  $dS_{\text{isolated}} \geq 0$ : **the entropy of an isolated system can never decrease; it increases in every irreversible process and remains constant only in a reversible one.** A system together with its surroundings forms an isolated system – the *universe* of the process – so for any process

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} = S_{\text{gen}} \geq 0 \quad (\text{principle of increase of entropy})$$

This is the most general statement of the second law. It gives the direction of processes (a process can occur only if it increases – or, ideally, leaves unchanged – the entropy of the universe), a quantitative measure of irreversibility (the larger  $S_{\text{gen}}$ , the more irreversible the process and the more work potential is destroyed, Section 16), and the criterion of equilibrium (an isolated system is in equilibrium when its entropy is a maximum). The surroundings are usually treated as a reservoir at  $T_0$  that exchanges heat reversibly, so  $\Delta S_{\text{surr}} = Q_{\text{surr}}/T_0$ , where  $Q_{\text{surr}}$  is the heat *received* by the surroundings.

**KEY POINT** Method for every "entropy of the universe" problem: (1) find  $\Delta S$  of the system along any reversible path between its end states ( $Tds$  equations, tables); (2) find the heat  $Q$  exchanged with the surroundings from the first law; (3)  $\Delta S_{\text{surr}} = -Q_{\text{sys}}/T_0$  if the surroundings are a reservoir at  $T_0$ ; (4)  $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} \geq 0$  – if it comes out negative, the process is impossible or an arithmetic error has been made.

## 11.3 Entropy generation in common irreversible processes

- **Heat transfer through a finite temperature difference.** Heat  $Q$  flows from a body at  $T_1$  to a body at  $T_2 < T_1$ :  $S_{\text{gen}} = -Q/T_1 + Q/T_2 = Q(T_1 - T_2)/(T_1 T_2) > 0$ . It vanishes only as  $T_1 \rightarrow T_2$  – reversible heat transfer needs an infinitesimal temperature difference. The larger the temperature difference across which the same heat is transferred, the greater the entropy generated (Worked Example 2.10, Figure 2.9).
- **Free (unrestrained) expansion of an ideal gas** from  $V_1$  to  $V_2$  in an insulated vessel:  $Q = 0$ ,  $W = 0$ ,  $T$  unchanged, so  $\Delta S_{\text{sys}} = mR \ln(V_2/V_1) > 0$  with  $\Delta S_{\text{surr}} = 0$ ;  $S_{\text{gen}} = mR \ln(V_2/V_1)$ .
- **Dissipation of work** (paddle wheel, electrical resistance, friction) in an insulated system: all the work becomes internal energy,  $Q = 0$ , and  $\Delta S = \int c \, dT/T > 0$  (Worked Example 2.7).
- **Mixing** of two masses of the same liquid at different temperatures  $T_a$  and  $T_b$  (insulated): the final temperature is the mass-weighted mean  $T_f$ , and  $S_{\text{gen}} = m_a c \ln(T_f/T_a) + m_b c \ln(T_f/T_b) > 0$ .
- **Melting of ice in warm surroundings, condensation of steam in a cold condenser, throttling** – all generate entropy because heat crosses a finite temperature difference or pressure falls without doing work (Worked Example 2.9).

## 11.4 Worked examples on entropy

### WORKED EXAMPLE 2.7 — Paddle work on an insulated gas

**Problem.** An insulated rigid vessel of 3 m<sup>3</sup> contains 15 kg of nitrogen. A paddle wheel stirs the gas until its pressure rises from 3 bar to 6 bar. Find (i) the change in internal energy, (ii) the work done, (iii) the heat transferred and (iv) the change in entropy of the gas. For nitrogen  $R = 0.297$  kJ/kg K and  $c_v = 0.743$  kJ/kg K.

**Given:**  $V = 3$  m<sup>3</sup> (constant);  $m = 15$  kg;  $p_1 = 300$  kPa;  $p_2 = 600$  kPa;  $R = 0.297$  kJ/kg K;  $c_v = 0.743$  kJ/kg K; insulated  $\Rightarrow Q = 0$ .

**Formula:**  $T = pV/(mR)$ ;  $\Delta U = m c_v(T_2 - T_1)$ ; first law  $Q = \Delta U + W$ ;  $S_2 - S_1 = m c_v \ln(T_2/T_1)$  (constant volume).

**Solution.**

$T_1 = p_1 V / (mR) = 300 \times 3 / (15 \times 0.297) = 202.0$  K;  $T_2 = 600 \times 3 / (15 \times 0.297) = 404.0$  K (the temperature doubles with the pressure at constant volume).

(i)  $\Delta U = 15 \times 0.743 \times (404.0 - 202.0) = 2251.5$  kJ.

(ii)  $Q = 0$ , so  $W = -\Delta U = -2251.5$  kJ; the negative sign shows that 2251.5 kJ of work is done *on* the gas by the paddle wheel.

(iii)  $Q = 0$  (insulated vessel).

(iv)  $S_2 - S_1 = 15 \times 0.743 \times \ln(404.0/202.0) = 15 \times 0.743 \times 0.6931 = 7.725$  kJ/K.

**ANSWER**  $\Delta U = +2251.5$  kJ;  $W = -2251.5$  kJ (work input);  $Q = 0$ ;  $\Delta S = +7.725$  kJ/K. The entropy rises although no heat is transferred – the increase is entirely *generated* by the dissipation of the paddle work, an irreversible process ( $S_{\text{gen}} = 7.725$  kJ/K).

### WORKED EXAMPLE 2.8 — Entropy change of water heated to superheated steam

**Problem.** Calculate the change in entropy when 4 kg of water at 25 °C and atmospheric pressure is converted into steam at 350 °C, considering separately (i) the heating of the water to 100 °C, (ii) the evaporation at 100 °C and (iii) the superheating of the steam. Take  $c_{\text{water}} = 4.187$  kJ/kg K,  $h_{\text{fg}} = 2257$  kJ/kg at 100 °C, and for steam  $c_p = R(3.5 + 1.2 \times 10^{-3} T + 0.14 \times 10^{-6} T^2)$  kJ/kg K with  $T$  in kelvin and  $R = 0.462$  kJ/kg K.

**Given:**  $m = 4$  kg;  $T_1 = 298.15$  K;  $T_{\text{sat}} = 373.15$  K;  $T_3 = 623.15$  K;  $c_w = 4.187$  kJ/kg K;  $h_{\text{fg}} = 2257$  kJ/kg;  $R = 0.462$  kJ/kg K.

**Formula:** (i)  $\Delta S_1 = m c_w \ln(T_{\text{sat}}/T_1)$ ; (ii)  $\Delta S_2 = m h_{\text{fg}}/T_{\text{sat}}$ ; (iii)  $\Delta S_3 = m \int c_p dT/T = m R [3.5 \ln(T_3/T_{\text{sat}}) + 1.2 \times 10^{-3}(T_3 - T_{\text{sat}}) + 0.07 \times 10^{-6}(T_3^2 - T_{\text{sat}}^2)]$ .

**Solution.**

(i) Heating of water:  $\Delta S_1 = 4 \times 4.187 \times \ln(373.15/298.15) = 16.748 \times 0.2244 = 3.758$  kJ/K.

(ii) Evaporation:  $\Delta S_2 = 4 \times 2257/373.15 = 24.194$  kJ/K.

(iii) Superheating:  $3.5 \ln(623.15/373.15) = 3.5 \times 0.5128 = 1.7948$ ;  $1.2 \times 10^{-3} \times (623.15 - 373.15) = 0.3000$ ;  $0.07 \times 10^{-6} \times (623.15^2 - 373.15^2) = 0.07 \times 10^{-6} \times 249075 = 0.0174$ .

$\Delta S_3 = 4 \times 0.462 \times (1.7948 + 0.3000 + 0.0174) = 1.848 \times 2.1122 = 3.903$  kJ/K.

Total:  $\Delta S = 3.758 + 24.194 + 3.903 = 31.86$  kJ/K.

**ANSWER**  $\Delta S_{\text{heating}} = 3.76$  kJ/K;  $\Delta S_{\text{evaporation}} = 24.19$  kJ/K;  $\Delta S_{\text{superheating}} = 3.90$  kJ/K; total  $\Delta S = 31.86$  kJ/K. Three-quarters of the entropy increase occurs during evaporation, where a large latent heat is absorbed at a comparatively low temperature.

**WORKED EXAMPLE 2.9 — Melting of ice in the atmosphere and the entropy of the universe**

**Problem.** One kilogram of ice at  $-8\text{ }^{\circ}\text{C}$  is exposed to the atmosphere at  $30\text{ }^{\circ}\text{C}$ ; it melts and the water finally reaches  $30\text{ }^{\circ}\text{C}$ . Determine (i) the entropy change of the ice–water system, (ii) the entropy change of the atmosphere, (iii) the entropy increase of the universe and (iv) the minimum work needed to convert the water back into ice at  $-8\text{ }^{\circ}\text{C}$ . Take  $c_{\text{ice}} = 2.093\text{ kJ/kg K}$ ,  $c_{\text{water}} = 4.187\text{ kJ/kg K}$  and  $h_{\text{sf}} = 334.5\text{ kJ/kg}$ .

**Given:**  $m = 1\text{ kg}$ ; ice from  $265.15\text{ K}$  to  $273.15\text{ K}$ ; melting at  $273.15\text{ K}$ ; water from  $273.15\text{ K}$  to  $303.15\text{ K}$ ; atmosphere  $T_0 = 303.15\text{ K}$ .

**Formula:**  $\Delta S_{\text{sys}} = m c_{\text{ice}} \ln(273.15/265.15) + m h_{\text{sf}}/273.15 + m c_{\text{w}} \ln(303.15/273.15)$ ;  $Q = m[c_{\text{ice}} \times 8 + h_{\text{sf}} + c_{\text{w}} \times 30]$ ;  $\Delta S_{\text{atm}} = -Q/T_0$ ;  $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{atm}}$ ;  $W_{\text{min}} = T_0 \Delta S_{\text{univ}}$ .

**Solution.**

(i) Warming of ice:  $2.093 \times \ln(273.15/265.15) = 2.093 \times 0.02973 = 0.0622\text{ kJ/K}$ .

Melting:  $334.5/273.15 = 1.2246\text{ kJ/K}$ .

Warming of water:  $4.187 \times \ln(303.15/273.15) = 4.187 \times 0.1042 = 0.4363\text{ kJ/K}$ .

$\Delta S_{\text{sys}} = 0.0622 + 1.2246 + 0.4363 = 1.7231\text{ kJ/K}$ .

(ii) Heat taken from the atmosphere:  $Q = 2.093 \times 8 + 334.5 + 4.187 \times 30 = 16.74 + 334.5 + 125.61 = 476.85\text{ kJ}$ .

$\Delta S_{\text{atm}} = -476.85/303.15 = -1.5730\text{ kJ/K}$ .

(iii)  $\Delta S_{\text{univ}} = 1.7231 - 1.5730 = 0.1501\text{ kJ/K}$  (positive – the process is irreversible, as expected for heat transfer through a finite temperature difference).

(iv) To restore the ice a refrigerator must remove  $Q = 476.85\text{ kJ}$  from the water and reject  $Q + W$  to the atmosphere. For the least work the refrigerator is reversible and the entropy of the universe is unchanged:  $-1.7231 + (476.85 + W_{\text{min}})/303.15 = 0$ , so  $W_{\text{min}} = 303.15 \times 1.7231 - 476.85 = 522.36 - 476.85 = 45.5\text{ kJ}$ . Equivalently  $W_{\text{min}} = T_0 \Delta S_{\text{univ}} = 303.15 \times 0.1501 = 45.5\text{ kJ}$  – the work lost in the spontaneous melting.

**ANSWER** (i)  $\Delta S_{\text{sys}} = +1.723\text{ kJ/K}$ ; (ii)  $\Delta S_{\text{atm}} = -1.573\text{ kJ/K}$ ; (iii)  $\Delta S_{\text{univ}} = +0.150\text{ kJ/K}$ ; (iv)  $W_{\text{min}} = 45.5\text{ kJ}$ .

**WORKED EXAMPLE 2.10 — Heating water from a reservoir – the effect of staging**

**Problem.** One kilogram of water at  $283\text{ K}$  is brought into contact with a heat reservoir at  $383\text{ K}$ . (i) When the water reaches  $383\text{ K}$ , find the entropy change of the water, of the reservoir and of the universe. (ii) If instead the water is first heated to  $333\text{ K}$  with a reservoir at  $333\text{ K}$  and then to  $383\text{ K}$  with the reservoir at  $383\text{ K}$ , find the entropy change of the universe. (iii) Comment on the effect of staging and explain how the water could be heated with negligible entropy generation. Take  $c_p = 4.187\text{ kJ/kg K}$ .

**Given:**  $m = 1\text{ kg}$ ;  $T_1 = 283\text{ K}$ ;  $T_2 = 383\text{ K}$ ; intermediate reservoir  $333\text{ K}$ ;  $c = 4.187\text{ kJ/kg K}$ .

**Formula:**  $\Delta S_{\text{water}} = m c \ln(T_2/T_1)$  (a property – the same in both cases);  $\Delta S_{\text{reservoir}} = -Q/T_{\text{res}}$  with  $Q = m c \Delta T$ ;  $\Delta S_{\text{univ}} = \Delta S_{\text{water}} + \sum \Delta S_{\text{res}}$ .

**Solution.**

(i)  $\Delta S_{\text{water}} = 4.187 \times \ln(383/283) = 4.187 \times 0.3026 = 1.2669\text{ kJ/K}$ .

Heat supplied by the reservoir  $Q = 4.187 \times (383 - 283) = 418.7\text{ kJ}$ ;  $\Delta S_{\text{res}} = -418.7/383 = -1.0932\text{ kJ/K}$ .

$\Delta S_{\text{univ}} = 1.2669 - 1.0932 = 0.1737\text{ kJ/K}$ .

(ii) Water: the same  $1.2669\text{ kJ/K}$ . First reservoir supplies  $4.187 \times 50 = 209.35\text{ kJ}$  at  $333\text{ K}$ :  $\Delta S = -209.35/333 = -0.6287\text{ kJ/K}$ ; second reservoir supplies  $209.35\text{ kJ}$  at  $383\text{ K}$ :  $\Delta S = -209.35/383 = -0.5466\text{ kJ/K}$ .

$\Delta S_{\text{univ}} = 1.2669 - 0.6287 - 0.5466 = 0.0917\text{ kJ/K}$ .

(iii) Staging reduces the entropy generation from 0.1737 to 0.0917 kJ/K, a reduction of 47 %, because each parcel of heat now crosses a smaller temperature difference (Figure 2.9). With  $n$  reservoirs equally spaced in temperature the generation falls further, and in the limit of an infinite number of reservoirs – each only infinitesimally hotter than the water at that moment – the heat transfer becomes reversible and  $\Delta S_{\text{univ}} \rightarrow 0$ . Regenerative feed heating in steam plants is the practical application of this idea.

**ANSWER** (i)  $\Delta S_{\text{water}} = +1.267$  kJ/K,  $\Delta S_{\text{res}} = -1.093$  kJ/K,  $\Delta S_{\text{univ}} = +0.174$  kJ/K; (ii)  $\Delta S_{\text{univ}} = +0.092$  kJ/K; (iii) staging nearly halves the entropy generation; an infinite sequence of reservoirs (or a reversible heat pump) would make it zero.

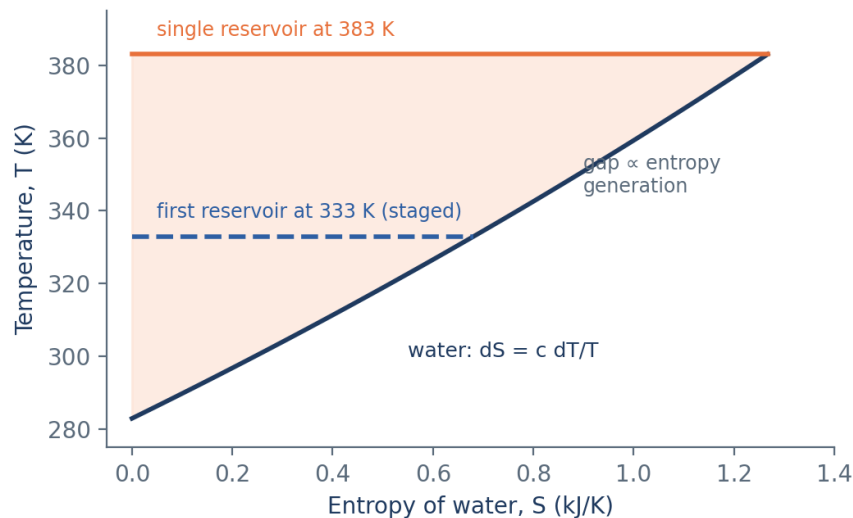


Figure 2.9 Heating water from 283 K to 383 K: the gap between the reservoir temperature and the water temperature is the source of entropy generation; a staged supply narrows the gap.

## 12. Applications of the second law

The second law is applied in every branch of engineering; the following are its principal uses, most of which are developed in this course.

- **Direction and feasibility of processes.** A proposed process or cycle is possible only if  $\Delta S_{\text{univ}} \geq 0$  (equivalently  $\oint \delta Q/T \leq 0$ ); this is the test used in Worked Examples 2.1 and 2.9 and in the analysis of every claimed invention.
- **Limits of performance.** The Carnot efficiency  $1 - T_2/T_1$  and the Carnot COPs are the ceilings for all power plants, engines, refrigerators and heat pumps; the designer compares an actual cycle (Rankine, Otto, Diesel, Brayton, vapour-compression) with these limits and knows how much improvement remains possible.
- **The absolute temperature scale.** The Kelvin scale (Section 7.3) is a consequence of the second law and is independent of any thermometric substance.
- **Entropy and the T-s diagram.** Entropy is the property that describes reversible adiabatic processes (isentropic expansion in turbines and nozzles, isentropic compression in compressors and pumps) and gives the reference against which real machines are rated through isentropic efficiencies.
- **Quantifying losses.** Entropy generation and irreversibility (Sections 15–16) measure the work destroyed by friction, throttling, mixing, combustion and heat transfer through finite temperature differences, and point to where a plant can be improved (second-law analysis, exergy accounting).
- **Property relations.** The Tds equations, and the Maxwell relations, Clausius–Clapeyron equation and Joule–Thomson coefficient derived from them (Unit IV), rest on entropy being a property.

- **Chemical and phase equilibrium, heat transfer and beyond.** The criteria of equilibrium (maximum entropy, minimum Gibbs function), the direction of heat flow, the impossibility of reaching absolute zero and the concept of the quality of energy all follow from the second law.

## 13. High-grade and low-grade energy

The first law treats all forms of energy as equivalent; the second law shows that they are not. Energy that can be converted *completely* into work – or from one form to another – is called **high-grade energy**; energy of which only a part can be converted into work, the fraction being limited by the second law, is called **low-grade energy**.

| High-grade energy                                      | Low-grade energy                                                   |
|--------------------------------------------------------|--------------------------------------------------------------------|
| Mechanical work; electrical energy                     | Heat or thermal energy                                             |
| Water power (hydraulic head), wind power, tidal power  | Heat from the combustion of fuels (coal, oil, gas, wood)           |
| Kinetic energy of a jet; energy of a rotating flywheel | Heat from nuclear fission or fusion                                |
| Animal and muscular work                               | Solar heat, geothermal heat, ocean thermal energy                  |
| Complete conversion possible (in principle to 100 %)   | Only a fraction $1 - T_0/T$ convertible; the rest must be rejected |

High-grade energy can be degraded completely into low-grade energy – the work of a brake becomes heat, an electric heater converts all its input into heat – but the reverse conversion is limited. The bulk of the high-grade energy used by society (electricity, shaft power) is obtained from low-grade energy (the heat of burning fuels), and the conversion is never complete: even the best steam power plant converts only about 40 to 45 % of the fuel heat into electricity. The part of the low-grade energy that can be converted is its **available energy**; the part that cannot is its **unavailable energy**. Because the second law places a limit on this conversion, engineering effort is directed at making the conversion as close to the limit as possible and at avoiding the needless degradation of high-grade energy (for example, using a heat pump instead of a resistance heater to warm a room).

## 14. Available and unavailable energy

### 14.1 Available energy of heat from a reservoir

Consider heat  $Q$  supplied by a reservoir at temperature  $T_1$  when the lowest available sink is the surroundings at  $T_0$ . The maximum work obtainable from  $Q$  is that of a Carnot engine between  $T_1$  and  $T_0$ :

$$AE = W_{\max} = Q(1 - T_0/T_1) \quad UE = Q - AE = Q T_0/T_1 = T_0 \Delta S$$

The **available energy (AE)**, also called the *available part of the energy* or *exergy*, is the portion of the heat that can be converted into useful work by an ideal engine; the **unavailable energy (UE)** is the portion that must be rejected to the surroundings and can never be converted into work. On the  $T$ - $s$  diagram (Figure 2.8) the heat  $Q$  is the whole area under the constant-temperature line  $T_1$  between  $s_1$  and  $s_2$ ; the unavailable energy is the rectangle  $T_0(s_2 - s_1)$  beneath  $T_0$ , and the available energy is the area between the two lines. The unavailable energy therefore depends on the temperature of the surroundings – on a cold day more of a given quantity of heat is available – and increases with the entropy transferred: **UE =  $T_0 \times$  (entropy change during heat supply)**.

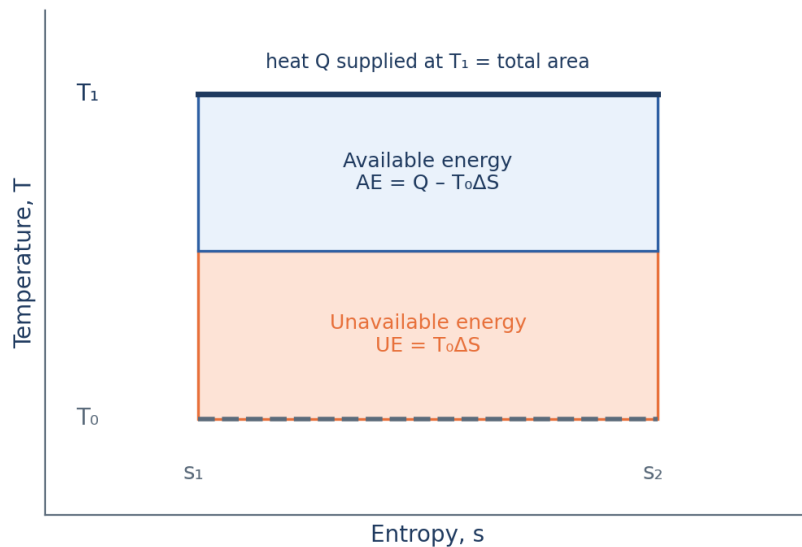


Figure 2.8 Available and unavailable energy of heat supplied at  $T_1$  with surroundings at  $T_0$  on the  $T$ - $s$  diagram.

## 14.2 Available energy of heat from a finite body

If the heat is not supplied at constant temperature but by a body (a stream of hot gas, for example) whose temperature falls from  $T_1$  to  $T_2$  as it gives up heat, the heat can be regarded as supplied in elements  $\delta Q$  at successively lower temperatures  $T$ , each driving an elementary Carnot engine:  $\delta W_{\max} = \delta Q(1 - T_0/T)$ . Integrating,

$$AE = Q - T_0 \int \delta Q/T = Q - T_0(S_1 - S_2)_{\text{body}} \quad UE = T_0(S_1 - S_2)_{\text{body}}$$

For a gas cooled at constant pressure from  $T_1$  to  $T_2$ ,  $Q = mc_p(T_1 - T_2)$  and  $UE = T_0 mc_p \ln(T_1/T_2)$ .

## 14.3 Loss of available energy due to heat transfer through a finite temperature difference

Let heat  $Q$  pass from a hot fluid at  $T_1$  to a cooler fluid at  $T_1' (< T_1)$  in a heat exchanger. Before the transfer the available energy of  $Q$  was  $Q - T_0Q/T_1$ ; after it, at the lower temperature, it is  $Q - T_0Q/T_1'$ . The loss is

$$\text{Loss of AE} = T_0(Q/T_1' - Q/T_1) = T_0 \Delta S_{\text{univ}} \quad (\text{greater unavailable energy after transfer})$$

The same heat delivered at a lower temperature carries more entropy and hence more unavailable energy. Every heat transfer across a finite temperature difference – in boilers, condensers, feed heaters – therefore *destroys* available energy, and the destruction is proportional to the entropy generated (Worked Example 2.10). Heat should be transferred across the smallest practicable temperature difference, which is the thermodynamic reason for regenerative feed heating, counter-flow heat exchangers and staged compression with intercooling.

# 15. Availability of closed systems and of steady-flow streams

## 15.1 The dead state and the definition of availability

The **dead state** is the state of a system when it is in complete (thermal, mechanical and chemical) equilibrium with the surroundings: at the pressure  $p_0$  and temperature  $T_0$  of the atmosphere (taken as 100 kPa and 25 °C unless otherwise stated), at rest, at the datum elevation and chemically inert. A system in the dead state has no capacity to do work. The **availability** (also called *exergy* or *available energy of a system*) is **the maximum useful work that can be obtained from a system in a given state as it is brought reversibly to the dead state, exchanging heat only with the surroundings**. Availability is a

property of the system *and* of the surroundings together – the same steam has more availability on a cold day.

Two qualifications are essential. First, the work must be *useful*: when a closed system expands, part of the work it does is spent in pushing back the atmosphere,  $p_0(V_2 - V_1)$ , and this part is not available for any useful purpose, so  $W_{\text{useful}} = W - p_0(V_2 - V_1)$ . Second, the process must be *reversible* and heat must be exchanged only with the surroundings at  $T_0$ ; heat exchanged with any other reservoir would itself be a source of work.

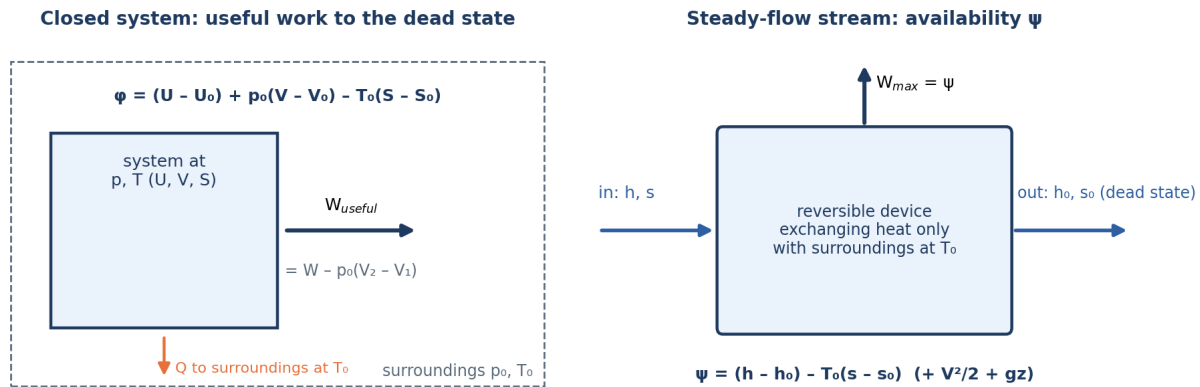


Figure 2.10 (Left) Useful work of a closed system going to the dead state. (Right) Availability of a steady-flow stream: the maximum work of a reversible device taking the stream to the dead state.

## 15.2 Availability of a closed system

Consider a closed system that changes from state 1 to state 2 while exchanging heat  $Q$  only with the surroundings at  $T_0$  and doing work  $W$  (Figure 2.10, left). The first law gives  $Q - W = U_2 - U_1$ . The entropy principle applied to the system plus surroundings gives  $(S_2 - S_1) + (-Q/T_0) \geq 0$ , i.e.  $Q \leq T_0(S_2 - S_1)$ . Eliminating  $Q$ ,

$$W = (U_1 - U_2) + Q \leq (U_1 - U_2) - T_0(S_1 - S_2).$$

The equality holds for a reversible process, so the maximum work is  $W_{\text{max}} = (U_1 - U_2) - T_0(S_1 - S_2)$ . Deducting the work done on the atmosphere gives the maximum *useful* work:

$$W_{\text{max,useful}} = (U_1 - U_2) + p_0(V_1 - V_2) - T_0(S_1 - S_2) = A_1 - A_2, \text{ where } A = U + p_0V - T_0S$$

$A$  is the **availability function of a closed system** (a combination of system properties and the surroundings constants  $p_0$  and  $T_0$ ). When the final state is the dead state, the maximum useful work is the **availability of the closed system**:

$$\phi = (U - U_0) + p_0(V - V_0) - T_0(S - S_0) \quad [\text{kJ}] \quad (\phi = A - A_0)$$

For a process between two states the **change in availability** is  $\phi_1 - \phi_2 = A_1 - A_2 = W_{\text{max,useful}}$ ; it differs from  $W_{\text{max}}$  by the atmospheric work  $p_0(V_2 - V_1)$  (Worked Example 2.11). If the system also possesses kinetic and potential energy,  $\frac{1}{2}mV^2 + mgz$  is added, since these are fully convertible into work.

## 15.3 Availability of a steady-flow stream

For a steady-flow device receiving fluid at state 1 and discharging it at state 2, exchanging heat  $q$  (per kg) only with the surroundings at  $T_0$  (Figure 2.10, right), the SFEE gives  $q - w = (h_2 - h_1) + (V_2^2 - V_1^2)/2 + g(z_2 - z_1)$ , and the entropy principle gives  $(s_2 - s_1) - q/T_0 \geq 0$ , i.e.  $q \leq T_0(s_2 - s_1)$ . Eliminating  $q$ ,

$$w_{\text{max}} = (h_1 - h_2) - T_0(s_1 - s_2) + (V_1^2 - V_2^2)/2 + g(z_1 - z_2) = b_1 - b_2, \text{ where } b = h - T_0s (+ V^2/2 + gz)$$

$\psi$  is the **availability function of a steady-flow stream** (the *Keenan function* or *Darrieus function*). No atmospheric-work correction is needed in steady flow, because the flow work is already included in the enthalpy. With the exit at the dead state ( $h_0$ ,  $s_0$ , at rest, at datum level) the maximum work is the **availability (exergy) of the stream**:

$$\psi = (h - h_0) - T_0(s - s_0) + V^2/2 + gz \quad [\text{kJ/kg}]$$

The maximum work of any steady-flow process is then  $w_{\max} = \psi_1 - \psi_2$ , the *decrease in availability* of the stream (Worked Example 2.12). For an ideal gas with constant specific heats,  $\psi = c_p(T - T_0) - T_0[c_p \ln(T/T_0) - R \ln(p/p_0)] + V^2/2 + gz$ . Note that  $\psi$  can be positive even when  $T < T_0$  (a cold stream can produce work through an engine that receives heat from the atmosphere) and that a stream at  $p > p_0$  has availability even at  $T = T_0$ .

## 15.4 Availability of heat and of work

For completeness: the availability of heat  $Q$  received from a reservoir at  $T$  is  $Q(1 - T_0/T)$  (Section 14.1), and the availability of work is the work itself – work is pure exergy. A quantity of heat at the temperature of the surroundings has zero availability.

# 16. Irreversibility and the Gouy–Stodola theorem

## 16.1 Definition

The **irreversibility**  $I$  of a process is the difference between the maximum useful work that could have been obtained (the reversible work) and the useful work actually obtained:

$$I = W_{\max, \text{useful}} - W_{\text{useful}} = W_{\text{rev}} - W_{\text{actual}} \quad (I \geq 0; I = 0 \text{ only for a reversible process})$$

It is the work *lost* – or, more precisely, the work potential *destroyed* – on account of the irreversibilities of the process; it is also called the *exergy destruction* or *lost work*. For a work-absorbing device (compressor, pump) it is the excess of the actual work input over the minimum (reversible) work input.

## 16.2 The Gouy–Stodola theorem

**Closed system.** From Section 15.2,  $W_{\max} = (U_1 - U_2) - T_0(S_1 - S_2)$  and, by the first law,  $W_{\text{actual}} = (U_1 - U_2) + Q$ , where  $Q$  is the heat received from the surroundings. Subtracting,  $I = W_{\max} - W_{\text{actual}} = T_0(S_2 - S_1) - Q = T_0[(S_2 - S_1) + (-Q/T_0)] = T_0(\Delta S_{\text{sys}} + \Delta S_{\text{surr}})$ .

**Steady-flow system.** Similarly, from Section 15.3,  $W_{\max} - W_{\text{actual}} = T_0(s_2 - s_1) - q = T_0[(s_2 - s_1)_{\text{stream}} + (-q/T_0)_{\text{surr}}]$ .

In both cases:

$$I = T_0 \Delta S_{\text{universe}} = T_0 S_{\text{gen}} \quad (\text{Gouy–Stodola theorem})$$

**The irreversibility of any process equals the temperature of the surroundings multiplied by the entropy generated in the process.** This elegant result links the two halves of the unit: the entropy calculations of Sections 9 to 11 tell us directly, through  $T_0$ , how much work potential each irreversible process destroys. In Worked Example 2.9 the melting of the ice generated 0.150 kJ/K, and  $T_0 \times 0.150 = 45.5$  kJ was the minimum work needed to undo it; in Worked Example 2.11 the entropy generated in the adiabatic expansion, multiplied by  $T_0$ , is the irreversibility 7.68 kJ. For a heat transfer  $Q$  from  $T_1$  to  $T_2$ ,  $I = T_0 Q(1/T_2 - 1/T_1)$ , which is the loss of available energy of Section 14.3.

## 17. First-law and second-law efficiency

### 17.1 First-law efficiency

The **first-law efficiency** (or energy efficiency) is the ratio of the desired energy output to the energy input, both measured simply as quantities of energy irrespective of their quality: the thermal efficiency of an engine,  $\eta_I = W/Q_1$ ; the COP of a refrigerator or heat pump; the efficiency of a boiler or a heater,  $\eta_I = \text{heat delivered}/\text{energy supplied}$ . Being based on the first law, it treats one kilojoule of heat at 100 °C as equal to one kilojoule of work and cannot show how close a device is to the best that the second law permits – a resistance heater has a first-law efficiency of 100 % and is nevertheless a thermodynamically poor way to heat a room.

### 17.2 Second-law efficiency

The **second-law efficiency** (effectiveness, exergetic efficiency) compares the actual performance with the best possible performance under the same conditions – the reversible performance. In terms of availability,

$$\eta_{II} = (\text{useful work or exergy actually obtained}) / (\text{maximum useful work or exergy supplied}) = W_{\text{useful}}/W_{\text{max}} = 1 - I/W_{\text{max}}$$

For the common devices:

| Device                                                          | First-law efficiency $\eta_I$              | Second-law efficiency $\eta_{II}$                                                   |
|-----------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------|
| Heat engine                                                     | $\eta_{th} = W/Q_1$                        | $\eta_{II} = \eta_{th}/\eta_{Carnot} = W/W_{\text{max}}$                            |
| Refrigerator or heat pump                                       | $\text{COP} = Q_2/W$ or $Q_1/W$            | $\eta_{II} = \text{COP}/\text{COP}_{\text{rev}} = W_{\text{min}}/W_{\text{actual}}$ |
| Turbine (steady flow)                                           | isentropic efficiency = $w/(h_1 - h_{2s})$ | $\eta_{II} = w_{\text{actual}}/(\psi_1 - \psi_2) = 1 - I/(\psi_1 - \psi_2)$         |
| Compressor or pump                                              | isentropic efficiency = $(h_{2s} - h_1)/w$ | $\eta_{II} = (\psi_2 - \psi_1)/w_{\text{actual}}$                                   |
| Heater supplying heat $Q$ at $T$ from work $W$ (or fuel exergy) | $Q/W$ ( $= 1$ for a resistance heater)     | $\eta_{II} = Q(1 - T_0/T)/W$                                                        |
| Heat exchanger                                                  | energy balance ( $\approx 1$ )             | $\eta_{II} = \text{exergy gained by cold stream}/\text{exergy lost by hot stream}$  |

The second-law efficiency is always between 0 and 1, and it identifies the real losses. A steam power plant with  $\eta_I = 35\%$  between 800 K and 300 K has  $\eta_{II} = 0.35/0.625 = 56\%$ ; a resistance heater delivering heat at 30 °C into a room when the surroundings are at 5 °C has  $\eta_{II} = (1 - 278.15/303.15) = 8.2\%$ , which tells us that a heat pump could do the same duty with a small fraction of the electricity. Worked Example 2.12 evaluates the second-law efficiency of a helium turbine.

### 17.3 Worked examples on availability and irreversibility

#### WORKED EXAMPLE 2.11 — Maximum work, change in availability and irreversibility of an adiabatic expansion

**Problem.** Two kilograms of air at 600 kPa and 100 °C expand adiabatically in a closed system until the volume is doubled and the temperature equals that of the surroundings, which are at 100 kPa and 15 °C. Determine (i) the maximum work, (ii) the change in availability and (iii) the irreversibility, and explain why the change in availability differs from the maximum work. Take  $c_v = 0.718$  kJ/kg K and  $R = 0.287$  kJ/kg K.

**Given:**  $m = 2$  kg;  $p_1 = 600$  kPa,  $T_1 = 373.15$  K;  $V_2 = 2V_1$ ,  $T_2 = T_0 = 288.15$  K;  $p_0 = 100$  kPa;  $c_v = 0.718$  kJ/kg K;  $R = 0.287$  kJ/kg K;  $Q = 0$ .

**Formula:**  $V = mRT/p$ ;  $W_{\max} = (U_1 - U_2) - T_0(S_1 - S_2)$ ;  $\phi_1 - \phi_2 = (U_1 - U_2) + p_0(V_1 - V_2) - T_0(S_1 - S_2)$ ;  $S_2 - S_1 = m[c_v \ln(T_2/T_1) + R \ln(V_2/V_1)]$ ;  $W_{\text{actual}} = U_1 - U_2$  (adiabatic);  $I = W_{\max} - W_{\text{actual}} = T_0 S_{\text{gen}}$ .

**Solution.**

$V_1 = 2 \times 0.287 \times 373.15/600 = 0.357 \text{ m}^3$ ;  $V_2 = 0.714 \text{ m}^3$ ;  $p_2 = 2 \times 0.287 \times 288.15/0.714 = 231.7 \text{ kPa}$ .

$U_1 - U_2 = m c_v(T_1 - T_2) = 2 \times 0.718 \times (373.15 - 288.15) = 122.06 \text{ kJ}$ .

$S_2 - S_1 = 2 \times [0.718 \times \ln(288.15/373.15) + 0.287 \times \ln 2] = 2 \times [0.718 \times (-0.2585) + 0.287 \times 0.6931] = 2 \times [-0.1856 + 0.1989] = 2 \times 0.01333 = 0.0267 \text{ kJ/K}$ .

(i)  $W_{\max} = 122.06 - 288.15 \times (-0.0267) = 122.06 + 7.68 = 129.74 \text{ kJ}$ .

(ii) Change in availability  $\phi_1 - \phi_2 = W_{\max} + p_0(V_1 - V_2) = 129.74 - 100 \times (0.714 - 0.357) = 129.74 - 35.70 = 94.04 \text{ kJ}$ .

(iii) Actual work of the adiabatic closed process:  $W = U_1 - U_2 = 122.06 \text{ kJ}$ . Irreversibility  $I = W_{\max} - W = 129.74 - 122.06 = 7.68 \text{ kJ}$ . Check:  $I = T_0 S_{\text{gen}} = T_0(S_2 - S_1)$  (since  $Q = 0$ )  $= 288.15 \times 0.0267 = 7.68 \text{ kJ} \checkmark$ .

**ANSWER** (i)  $W_{\max} = 129.7 \text{ kJ}$ ; (ii)  $\phi_1 - \phi_2 = 94.0 \text{ kJ}$ ; (iii)  $I = 7.68 \text{ kJ}$ . The change in availability is smaller than the maximum work by  $p_0(V_2 - V_1) = 35.7 \text{ kJ}$ , the work the expanding air must do merely to push back the atmosphere – work which is done in any case and is not useful. The irreversibility arises because the actual expansion is not reversible: its entropy increases (0.0267 kJ/K) although no heat is transferred.

**WORKED EXAMPLE 2.12 — Availability, irreversibility and second-law efficiency of a helium turbine**

**Problem.** Helium enters a turbine at 250 kPa and 250 °C and leaves at 100 kPa and 120 °C; the heat lost to the atmosphere, which is at 100 kPa and 25 °C, is 5 kJ/kg. Calculate (i) the availability of the entering stream, (ii) the availability of the leaving stream, (iii) the maximum work, (iv) the actual work, (v) the irreversibility and (vi) the second-law efficiency of the turbine. For helium  $c_p = 5.2 \text{ kJ/kg K}$  and molar mass = 4.003 kg/kmol. Neglect kinetic and potential energy changes.

**Given:**  $p_1 = 250 \text{ kPa}$ ,  $T_1 = 523.15 \text{ K}$ ;  $p_2 = 100 \text{ kPa}$ ,  $T_2 = 393.15 \text{ K}$ ;  $p_0 = 100 \text{ kPa}$ ,  $T_0 = 298.15 \text{ K}$ ;  $q_{\text{loss}} = 5 \text{ kJ/kg}$ ;  $c_p = 5.2 \text{ kJ/kg K}$ ;  $R = 8.3145/4.003 = 2.077 \text{ kJ/kg K}$ .

**Formula:**  $\psi = (h - h_0) - T_0(s - s_0) = c_p(T - T_0) - T_0[c_p \ln(T/T_0) - R \ln(p/p_0)]$ ;  $w_{\max} = \psi_1 - \psi_2$ ;  $w_{\text{actual}} = (h_1 - h_2) - q_{\text{loss}} = c_p(T_1 - T_2) - q_{\text{loss}}$ ;  $I = w_{\max} - w_{\text{actual}} = T_0 S_{\text{gen}}$ ;  $\eta_{II} = w_{\text{actual}}/w_{\max}$ .

**Solution.**

(i) Entering stream:  $h_1 - h_0 = 5.2 \times (523.15 - 298.15) = 1170.0 \text{ kJ/kg}$ .

$s_1 - s_0 = 5.2 \ln(523.15/298.15) - 2.077 \ln(250/100) = 5.2 \times 0.5623 - 2.077 \times 0.9163 = 2.9238 - 1.9031 = 1.0207 \text{ kJ/kg K}$ .

$\psi_1 = 1170.0 - 298.15 \times 1.0207 = 1170.0 - 304.3 = 865.7 \text{ kJ/kg}$ .

(ii) Leaving stream:  $h_2 - h_0 = 5.2 \times (393.15 - 298.15) = 494.0 \text{ kJ/kg}$ ;  $s_2 - s_0 = 5.2 \ln(393.15/298.15) - 2.077 \ln(100/100) = 5.2 \times 0.2766 - 0 = 1.4383 \text{ kJ/kg K}$ .

$\psi_2 = 494.0 - 298.15 \times 1.4383 = 494.0 - 428.8 = 65.2 \text{ kJ/kg}$ .

(iii)  $w_{\max} = \psi_1 - \psi_2 = 865.7 - 65.2 = 800.5 \text{ kJ/kg}$ .

(iv)  $w_{\text{actual}} = 5.2 \times (523.15 - 393.15) - 5 = 676.0 - 5 = 671.0 \text{ kJ/kg}$ .

(v)  $I = 800.5 - 671.0 = 129.5 \text{ kJ/kg}$ . Check by the Gouy–Stodola theorem:  $s_{\text{gen}} = (s_2 - s_1) + q_{\text{loss}}/T_0 = (1.4383 - 1.0207) + 5/298.15 = 0.4176 + 0.0168 = 0.4344 \text{ kJ/kg K}$ ;  $T_0 s_{\text{gen}} = 298.15 \times 0.4344 = 129.5 \text{ kJ/kg} \checkmark$ .

(vi)  $\eta_{II} = 671.0/800.5 = 0.838$ .

**ANSWER** (i)  $\psi_1 = 865.7 \text{ kJ/kg}$ ; (ii)  $\psi_2 = 65.2 \text{ kJ/kg}$ ; (iii)  $w_{\max} = 800.5 \text{ kJ/kg}$ ; (iv)  $w_{\text{actual}} = 671.0 \text{ kJ/kg}$ ; (v)  $I = 129.5 \text{ kJ/kg}$ ; (vi)  $\eta_{II} = 83.8 \%$ . The turbine recovers 84 % of the exergy given up by the helium; the remaining 16 % is destroyed – mostly by internal friction (the entropy rises by 0.418 kJ/kg K within the turbine) and to a small extent by the heat leak to the atmosphere.

## 18. Practice problems (answers given)

| No. | Problem                                                                                                                                                                                                                                      | Answer                                                                              |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| P1  | A Carnot engine receives 100 kW from a source at 500 °C and rejects heat to a sink at 30 °C. Find the power developed and the rate of heat rejection.                                                                                        | $\eta = 0.608$ ; $W = 60.8$ kW; $Q_2 = 39.2$ kW                                     |
| P2  | A reversible refrigerator keeps a cold room at -10 °C and rejects heat to surroundings at 35 °C. If 5 kW of heat leaks into the room, find the least power required.                                                                         | $COP = 5.85$ ; $W = 0.855$ kW                                                       |
| P3  | A heat pump with a COP of 3.5 delivers 14 kW to a building. Find the power input and the heat absorbed from the outside air.                                                                                                                 | $W = 4$ kW; $Q_2 = 10$ kW                                                           |
| P4  | An inventor claims an engine that receives 1200 kJ from a reservoir at 600 K and rejects 500 kJ to a reservoir at 300 K. Is the claim valid?                                                                                                 | $\eta_{\text{claimed}} = 0.583 > \eta_{\text{Carnot}} = 0.5$ ; impossible           |
| P5  | Two kilograms of air are heated at constant pressure from 300 K to 600 K. Find the entropy change ( $c_p = 1.005$ kJ/kg K).                                                                                                                  | $\Delta S = 1.393$ kJ/K                                                             |
| P6  | Equal masses (0.5 kg each) of water at 20 °C and 80 °C are mixed in an insulated vessel. Find the final temperature and the entropy generated.                                                                                               | 50 °C; $S_{\text{gen}} = 0.0181$ kJ/K                                               |
| P7  | Three kilograms of an ideal gas ( $R = 0.287$ kJ/kg K) expand freely in an insulated vessel from 1 m <sup>3</sup> to 3 m <sup>3</sup> . Find the entropy change of the gas and of the universe.                                              | $\Delta S = 0.946$ kJ/K (both)                                                      |
| P8  | A quantity of 1500 kJ of heat is transferred from a reservoir at 800 K to one at 400 K; the surroundings are at 300 K. Find the entropy generated and the loss of available energy.                                                          | $S_{\text{gen}} = 1.875$ kJ/K; loss = 562.5 kJ (AE falls from 937.5 to 375 kJ)      |
| P9  | Steam enters a turbine with $h = 3200$ kJ/kg, $s = 6.8$ kJ/kg K and leaves with $h = 2600$ kJ/kg, $s = 7.0$ kJ/kg K; heat loss 20 kJ/kg; $T_0 = 298$ K. Find the actual work, the maximum work, the irreversibility and $\eta_{\text{II}}$ . | $w = 580$ , $w_{\text{max}} = 659.6$ , $I = 79.6$ kJ/kg; $\eta_{\text{II}} = 0.879$ |
| P10 | A heat engine has a thermal efficiency of 35 % while working between 900 K and 300 K. Find its second-law efficiency.                                                                                                                        | $\eta_{\text{II}} = 0.35/0.667 = 0.525$                                             |

## 19. Summary and formula sheet

- Heat engine:**  $W = Q_1 - Q_2$ ,  $\eta = W/Q_1 = 1 - Q_2/Q_1$ . **Refrigerator:**  $(COP)_{\text{ref}} = Q_2/W$ . **Heat pump:**  $(COP)_{\text{HP}} = Q_1/W = (COP)_{\text{ref}} + 1 > 1$ .
- Kelvin–Planck:** no cyclic device can produce work while exchanging heat with a single reservoir (no PMM2). **Clausius:** no cyclic device can transfer heat from a cold to a hot body with no other effect. The two statements are equivalent.
- Carnot cycle:** two reversible isotherms + two reversible adiabatics;  $\eta_{\text{Carnot}} = 1 - T_2/T_1$ ; reversed cycle  $(COP)_{\text{ref}} = T_2/(T_1 - T_2)$ ,  $(COP)_{\text{HP}} = T_1/(T_1 - T_2)$ . Not practicable (infinitely slow isotherms, very small work per cycle, ideal insulation/conduction impossible). **Carnot theorem:**  $\eta_{\text{irrev}} < \eta_{\text{rev}}$ ; all reversible engines between the same reservoirs have the same efficiency, which depends only on  $T_1$  and  $T_2$ ;  $(Q_1/Q_2)_{\text{rev}} = T_1/T_2$  defines the Kelvin scale.
- Clausius inequality:**  $\oint \delta Q/T \leq 0$  ( $= 0$  reversible). **Entropy:**  $dS = (\delta Q/T)_{\text{rev}}$ , kJ/K; a property;  $Q_{\text{rev}} = \int T dS$  = area under the  $T$ – $s$  curve; isentropic = reversible adiabatic.
- Tds equations:**  $T ds = du + p dv = dh - v dp$ . **Ideal gas:**  $s_2 - s_1 = c_v \ln(T_2/T_1) + R \ln(v_2/v_1) = c_p \ln(T_2/T_1) - R \ln(p_2/p_1)$ ; constant  $V$ :  $mc_v \ln(T_2/T_1)$ ; constant  $p$ :  $mc_p \ln(T_2/T_1)$ ; isothermal:  $mR \ln(V_2/V_1) = Q/T$ ; polytropic:  $mc_v[(n - \gamma)/(n - 1)] \ln(T_2/T_1)$ . **Liquids/solids:**  $mc \ln(T_2/T_1)$ ; **phase change:**  $m h_{\text{fg}}/T_{\text{sat}}$ .

- **Increase of entropy:**  $dS \geq \delta Q/T$ ;  $S_2 - S_1 = \int \delta Q/T + S_{\text{gen}}$ ,  $S_{\text{gen}} \geq 0$ ;  $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} \geq 0$ ;  $\Delta S_{\text{surr}} = -Q_{\text{sys}}/T_0$  for a reservoir. Heat transfer across  $\Delta T$ :  $S_{\text{gen}} = Q(1/T_2 - 1/T_1)$ .
- **High-grade energy** (work, electricity, hydraulic, wind) is fully convertible; **low-grade energy** (heat from fuels, nuclear, solar) only partly. **Available energy** of heat  $Q$  at  $T_1$ :  $AE = Q(1 - T_0/T_1)$ ; **unavailable energy**  $UE = T_0\Delta S$ ; loss of AE in heat transfer  $= T_0\Delta S_{\text{univ}}$ .
- **Availability (exergy):** maximum useful work in going reversibly to the dead state ( $p_0, T_0$ ). Closed system:  $\phi = (U - U_0) + p_0(V - V_0) - T_0(S - S_0)$ ;  $W_{\text{max,useful}} = \phi_1 - \phi_2$ ;  $W_{\text{max}} = (U_1 - U_2) - T_0(S_1 - S_2)$ . Steady flow:  $\psi = (h - h_0) - T_0(s - s_0) + V^2/2 + gz$ ;  $w_{\text{max}} = \psi_1 - \psi_2$ .
- **Irreversibility:**  $I = W_{\text{max}} - W_{\text{actual}} = T_0 S_{\text{gen}} = T_0 \Delta S_{\text{univ}}$  (Gouy–Stodola). **First-law efficiency** = energy out/energy in; **second-law efficiency**  $\eta_{\text{II}} = W_{\text{actual}}/W_{\text{max}} = \eta_I/\eta_{\text{rev}}$  (engines) =  $\text{COP}/\text{COP}_{\text{rev}}$  (refrigerators) =  $1 - I/W_{\text{max}}$ .

## 20. Review questions

### 20.1 Short-answer (two-mark) questions

1. Write the Kelvin–Planck statement of the second law in your own words.
2. Write the Clausius statement of the second law.
3. What is a heat engine? Give the expression for its thermal efficiency in terms of the heat quantities.
4. What is a refrigerator? Define its coefficient of performance.
5. What is a heat pump? How is its COP defined?
6. In what way does a heat pump differ from a refrigerator when both are the same machine?
7. What is a thermal energy reservoir? When is a reservoir called a source and when a sink?
8. What is meant by a perpetual motion machine of the second kind? Which law does it violate?
9. List, in sequence, the four processes of the Carnot cycle.
10. Give the expression for the efficiency of a Carnot engine working between temperatures  $T_1$  and  $T_2$ .
11. State the Carnot theorem.
12. State any two corollaries of the Carnot theorem.
13. Mention two reasons why a Carnot engine cannot be built in practice.
14. Draw the reversed Carnot cycle on the  $T$ – $s$  plane and on the  $p$ – $V$  plane.
15. State the Clausius inequality and write it in mathematical form.
16. Define entropy and give its unit.
17. What is meant by the availability of a system?
18. Write the entropy change of an ideal gas between two states in terms of temperatures and volumes.
19. What is the entropy change of a system that receives heat  $Q$  reversibly at a constant temperature  $T$ ?
20. State the principle of increase of entropy.
21. What are available energy and unavailable energy?
22. How do high-grade and low-grade energy differ? Give two examples of each.
23. Why can the COP of a heat pump never be less than one?
24. Define the irreversibility of a process.
25. How does the second-law efficiency differ from the first-law efficiency?
26. What is the dead state?
27. Write the two Tds equations.
28. State the Gouy–Stodola theorem.
29. Why is the entropy change of an isolated system never negative?
30. Why is heat transfer through a finite temperature difference irreversible?

### 20.2 Long-answer (thirteen-mark) questions

1. Describe the Carnot cycle for a gas with  $p$ – $V$  and  $T$ – $s$  diagrams, derive its efficiency in terms of the reservoir temperatures, and state and prove the Carnot theorem and its corollaries.
2. With schematic diagrams explain the working of a heat engine, a refrigerator and a heat pump; define the performance parameter of each, derive the expressions and show that  $(\text{COP})_{\text{HP}} = (\text{COP})_{\text{ref}} + 1$ .

3. State the Kelvin–Planck and Clausius statements of the second law and prove that they are equivalent.
4. A cyclic heat engine works between a source at 750 °C and a sink at 25 °C. (i) Find the least heat rejection per kW of net work. (ii) An inventor claims an engine between the same temperatures producing 1 kW while rejecting 0.35 kW; examine the claim. (iii) A real engine rejects 0.7 kW per kW of output; find its efficiency and its ratio to the maximum.
5. Two reversible engines A and B are in series; A receives 250 kJ at 450 °C and B rejects heat at 10 °C; the work of A is twice that of B. Find the intermediate temperature, the efficiency of each engine and the heat rejected to the sink.
6. A reversible heat pump keeps a house at 22 °C; the heat transfer through the structure is 0.6 kW per kelvin of temperature difference. Find the minimum power when the outside is at 2 °C in winter, and the highest summer outside temperature for which the same power can hold 22 °C inside.
7. A reversible engine between 650 °C and 40 °C receives 2000 kJ and drives a reversible refrigerator between 40 °C and –20 °C; the net work of the plant is 300 kJ. Find the heat absorbed by the refrigerant and the net heat rejected to the 40 °C reservoir; repeat when the engine efficiency and refrigerator COP are each 50 % of their maximum values.
8. An ice plant on a reversed Carnot cycle makes 12 t of ice per day at 0 °C from water at 0 °C, rejecting heat to the atmosphere at 30 °C; it is driven by a Carnot engine receiving heat at 200 °C from a fuel of calorific value 42000 kJ/kg. Find the engine power and the fuel consumption per hour ( $h_{sf} = 334.5$  kJ/kg).
9. A freezer holds –18 °C in a room at 28 °C with a heat leak of 1.5 kW. Find the least power; if the compressor draws 0.6 kW find the actual COP, the heat rejected and the ratio of actual to maximum COP; and find the change in least power if the freezer is held at –28 °C.
10. Derive the Clausius inequality and, using it, prove that entropy is a property and that  $dS \geq \delta Q/T$  for any process.
11. An insulated vessel of 3 m<sup>3</sup> holds 15 kg of nitrogen ( $R = 0.297$ ,  $c_v = 0.743$  kJ/kg K); paddle work raises its pressure from 3 bar to 6 bar. Find the change in internal energy, the work, the heat transfer and the change in entropy.
12. Find the entropy change when 4 kg of water at 25 °C is converted to steam at 350 °C at atmospheric pressure, treating heating, evaporation and superheating separately ( $c_w = 4.187$  kJ/kg K,  $h_{fg} = 2257$  kJ/kg,  $c_{p,steam} = R(3.5 + 1.2 \times 10^{-3}T + 0.14 \times 10^{-6}T^2)$ ,  $R = 0.462$  kJ/kg K).
13. One kilogram of ice at –8 °C melts in the atmosphere at 30 °C and the water reaches 30 °C. Find the entropy changes of the system, of the atmosphere and of the universe, and the minimum work to re-freeze the water to –8 °C ( $c_{ice} = 2.093$ ,  $c_w = 4.187$  kJ/kg K,  $h_{sf} = 334.5$  kJ/kg).
14. One kilogram of water at 283 K is heated to 383 K by a reservoir at 383 K; find the entropy changes of the water, the reservoir and the universe. Repeat with reservoirs at 333 K and 383 K used in succession, and explain how the entropy generation could be made negligible.
15. Explain the terms dead state, availability of a closed system, availability of a steady-flow stream and irreversibility; derive the expressions for the maximum useful work of a closed system and of a steady-flow stream, and prove that  $I = T_0 S_{gen}$ .
16. Two kilograms of air at 600 kPa and 100 °C expand adiabatically in a closed system to twice the volume and to the surroundings temperature of 15 °C ( $p_0 = 100$  kPa). Find the maximum work, the change in availability and the irreversibility, and explain why the first two differ ( $c_v = 0.718$ ,  $R = 0.287$  kJ/kg K).
17. Helium ( $c_p = 5.2$  kJ/kg K,  $M = 4.003$  kg/kmol) enters a turbine at 250 kPa, 250 °C and leaves at 100 kPa, 120 °C, losing 5 kJ/kg of heat to the atmosphere at 100 kPa and 25 °C. Find the availability of the entering and leaving streams, the maximum work, the actual work, the irreversibility and the second-law efficiency.

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*Text: P. K. Nag, "Engineering Thermodynamics", 6th Edn., McGraw-Hill — Chapters 6–8. References: Cengel & Boles, "Thermodynamics: An Engineering Approach", 8th Edn.; R. K. Rajput, "A Text Book of Engineering Thermodynamics", 5th Edn.*